Machine Learning Moist Physics in DIMSWE: Effects of Physical Representation, Discrete Consistency, and Recursive Training

1 Introduction

We report the findings from a controlled set of machine-learning studies performed with DIMSWE, a moist shallow-water solver. The goal is to determine how different learning objectives affect the learned embedded physics and to isolate the contributions of deployment consistency and recursive model-state feedback that are often grouped together under the broad label of online training. In particular, deployment consistency accounts for the numerical discretization and the state at which the learned physics is evaluated within the deployed timestep, whereas genuine recursive feedback arises when the learned model is evaluated on states that have themselves been influenced by earlier model predictions. Exact training with such recursive model-state feedback generally requires propagation through the deployed solver, i.e. solver-in-the-loop training. Related LES work has demonstrated the importance of explicitly accounting for the deployed discretization and model–data consistency during closure learning, and has shown that full solver-in-the-loop training is not always necessary to achieve stable and accurate deployment in the settings considered [1]. Recursive solver feedback may also be approximated without differentiating through the complete numerical solver; for example, approximate online-gradient methods have been proposed for hybrid modeling systems [3].

In section 2, we begin with the continuous governing equations solved within DIMSWE, followed by a brief description of the numerical discretization and timestep. We then introduce three different machine-learning representations of the moist physics. We have also considered six different learning objectives, which are described next. The section concludes with the neural-network architecture and training details. In section 3, we describe the ground-truth numerical simulations and their physical evolution. The results obtained from the various combinations of learned-physics representations and training objectives are presented in section 4, primarily using machine-learning and process-level metrics. The corresponding hybrid simulations are examined from a physical and dynamical perspective in section 5. Finally, the main findings and broader implications are summarized in section 6.

2 Mathematical and Numerical Formulation

2.1 Governing equations and moist-physics parameterization of the DIMSWE model

DIMSWE solves a moist thermal shallow-water model in which the resolved shallow-water dynamics are coupled to a simplified representation of phase changes and conversions between water vapor, cloud water, and rain water. The formulation used here follows the moist shallow-water framework of [2]. The prognostic state is

$$X = (\mathbf{v},\, h,\, S,\, Q_v,\, Q_c,\, Q_r)\,, \tag{1}$$

where,

$$S = hb, \qquad Q_v = hq_v, \qquad Q_c = hq_c, \qquad Q_r = hq_r. \tag{2}$$

Here, $\mathbf{v} = (u, v)^T$ is the horizontal velocity, h is the fluid-layer depth, and $b = S/h$ is the buoyancy (thermal) variable. The quantities q_v, q_c, and q_r denote the specific vapor, cloud-water, and rain-water mixing ratios, respectively, while $Q_i = hq_i$ are the corresponding depth-integrated moisture variables. The bottom topography is denoted by B. In this report $B = 0$. The state is governed by

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{v} + f\mathcal{R}(\mathbf{v}) + \nabla B + b\nabla h + \frac{h}{2}\nabla b = -\mu\nabla^4\mathbf{v}, \tag{3}$$

$$\frac{\partial h}{\partial t} + \nabla\cdot(h\mathbf{v}) = -\mu\nabla^4 h, \tag{4}$$

$$\frac{\partial S}{\partial t} + \nabla\cdot(S\mathbf{v}) = -\mu\nabla^4 S + h\beta_2 A, \tag{5}$$

$$\frac{\partial Q_v}{\partial t} + \nabla\cdot(Q_v\mathbf{v}) = hA, \tag{6}$$

$$\frac{\partial Q_c}{\partial t} + \nabla\cdot(Q_c\mathbf{v}) = -h(A+R), \tag{7}$$

$$\frac{\partial Q_r}{\partial t} + \nabla\cdot(Q_r\mathbf{v}) = hR. \tag{8}$$

Here,

$$\mathcal{R}(\mathbf{v}) = \begin{bmatrix} -v \\ u \end{bmatrix}, \tag{9}$$

is the 90° rotation associated with the Coriolis effect. The biharmonic terms proportional to μ provide numerical hyperviscosity and parity with HOMME. Moist processes enter the thermal and moisture equations through two scalar rates. The signed vapor–cloud phase-change rate is

$$A = E - C, \tag{10}$$

where C and E denote condensation and evaporation, respectively. The cloud-to-rain conversion rate is

$$R = \max\left[0,\ \gamma_r \frac{q_c - q_{\text{precip}}}{\tau_r}\right]. \tag{11}$$

where q_{precip} is the prescribed cloud-water threshold for rain formation, τ_r is the prescribed cloud-to-rain conversion timescale and γ_r is the corresponding conversion coefficient.

The condensation and evaporation rates are given by

$$C = \max\left(0,\ \gamma_v \frac{q_v - q_{\text{sat}}}{\tau_v}\right), \tag{12}$$

$$E = \min\left[\frac{q_c}{\Delta t},\ \max\left(0,\ \gamma_v \frac{q_{\text{sat}} - q_v}{\tau_v}\right)\right]. \tag{13}$$

The vapor field is relaxed toward the local saturation state over the timescale τ_v. With the sign convention in (10), $A < 0$ corresponds to net condensation, and therefore conversion of water vapor into cloud liquid, whereas $A > 0$ corresponds to net evaporation and conversion of cloud liquid into water vapor. The saturation mixing ratio is

$$q_{\text{sat}} = q_0 \frac{H_0}{h+B} \exp\left[20\left(1 - \frac{b}{g}\right)\right], \tag{14}$$

and

$$\gamma_v = \frac{1}{1 + 20 q_{\text{sat}} \beta_2 / g}, \qquad \beta_2 = gL. \tag{15}$$

The upper bound $q_c/\Delta t$ in the evaporation rate prevents the evaporation step from removing more cloud water than is locally available during a single timestep. Thus, although the model is written above in continuum form, this limiter introduces an explicit dependence of the moist parameterization on the numerical timestep.

The rate R provides a simplified parameterization of the conversion of cloud liquid to rain liquid. It is nonnegative and becomes active only when the cloud-water mixing ratio exceeds the prescribed threshold q_{precip}. Condensation can contribute indirectly to rain formation by increasing q_c and thereby driving the system above this threshold. The detailed evolution of the droplet size distribution and the effects of condensation, turbulence, and collision–coalescence on the eventual formation and growth of rain are not resolved individually; their aggregate effect on the cloud-to-rain conversion is represented here by the threshold and conversion timescale in (11). The parameterization contains no reverse rain-to-cloud conversion and does not explicitly represent droplet breakup or other size-redistribution processes.

Importantly, the moist source terms in (6)–(8) exactly balance. Defining the total depth-integrated water content as

$$Q_w = Q_v + Q_c + Q_r, \tag{16}$$

and summing the three moisture equations gives

$$\frac{\partial Q_w}{\partial t} + \nabla \cdot (Q_w \mathbf{v}) = hA - h(A + R) + hR = 0. \tag{17}$$

Thus the moist processes only redistribute water among vapor, cloud liquid, and rain liquid; they do not create or destroy total water. For the doubly periodic 2D domains considered here, integration of (17) over the domain therefore yields global conservation of total water. In models containing precipitation fallout or open boundaries, the corresponding water fluxes would additionally enter the global budget. In addition to conserving total water, the prescribed moist source terms satisfy an exact coupling between the thermodynamic and vapor tendencies,

$$\left(\frac{\partial S}{\partial t}\right)_{\text{moist}} - \beta_2 \left(\frac{\partial Q_v}{\partial t}\right)_{\text{moist}} = 0. \tag{18}$$

This identity represents the prescribed thermodynamic response associated with vapor–cloud phase change and plays the role of latent-heating coupling in this reduced shallow-water model.

2.2 Numerical discretization and timestep

DIMSWE uses a compatible finite-element spatial discretization. The horizontal velocity, layer depth, and thermal variable are represented using continuous higher-order finite elements,

$$\mathbf{v} \in [CG_3]^2, \qquad h, S \in CG_3, \tag{19}$$

whereas the depth-integrated moisture variables are represented using discontinuous elements,

$$Q_v, Q_c, Q_r \in DG_1. \tag{20}$$

The moisture variables are transported using an upwind discontinuous-Galerkin scheme. We omit the detailed finite-element weak forms here and focus on the time integration, which is particularly important for the machine-learning objectives considered below.

One DIMSWE timestep is composed of four sequential pieces: evolution of the dry shallow-water dynamics ($\Phi^{\mathrm{RK4}}_{\mathrm{dry}}$), application of hyperviscosity (Φ^{E}_{HV}), transport of the moisture variables ($\Phi^{\mathrm{SSPRK43}}_{\mathrm{DG}}$), and finally application of the local moist-physics source ($\Phi^{E}_{\mathrm{moist}}$).

$$X^{n+1} = \Phi^{E}_{\mathrm{moist}}(\Delta t) \circ \left[\Phi^{\mathrm{SSPRK43}}_{\mathrm{DG}}\left(\frac{\Delta t}{2}\right)\right]^2 \circ \Phi^{E}_{\mathrm{HV}}(\Delta t) \circ \left[\Phi^{\mathrm{RK4}}_{\mathrm{dry}}\left(\frac{\Delta t}{2}\right)\right]^2 (X^n). \tag{21}$$

Here, the dry dynamics are advanced using two RK4 half steps, hyperviscosity is applied with a forward-Euler step, and moisture transport is advanced using two SSPRK(4,3) half steps. The moist source terms are then applied through a final forward-Euler update over Δt.

For the discussion of learned moist physics, it is useful to collect all operations preceding the moist-physics update into a prefix operator,

$$P = \left[\Phi^{\mathrm{SSPRK43}}_{\mathrm{DG}}\left(\frac{\Delta t}{2}\right)\right]^2 \circ \Phi^{E}_{\mathrm{HV}}(\Delta t) \circ \left[\Phi^{\mathrm{RK4}}_{\mathrm{dry}}\left(\frac{\Delta t}{2}\right)\right]^2. \tag{22}$$

The state immediately before evaluation of the moist physics is therefore

$$Y^n = P(X^n). \tag{23}$$

Thus, Y^n, rather than the timestep-boundary state X^n, is the state actually seen by the moist-physics parameterization. The overall time step is therefore schematically represented as

$$X^n \xrightarrow[\Delta t/2]{\text{Dry RK4}} \xrightarrow[\Delta t/2]{\text{Dry RK4}} \xrightarrow[\Delta t]{\text{Hypervisc. Euler}} \xrightarrow[\Delta t/2]{\text{DG SSPRK43}} \xrightarrow[\Delta t/2]{\text{DG SSPRK43}} Y^n \xrightarrow[\Delta t]{\text{Moist Euler}} X^{n+1}. \tag{24}$$

Each split component is integrated using the indicated step size, but these step sizes should not be summed to determine the elapsed physical time. The operators represent separate contributions to the DIMSWE tendency and are composed as a splitting approximation over a single physical timestep Δt. Thus, X^n and X^{n+1} correspond to t^n and $t^{n+1} = t^n + \Delta t$.

At Y^n, the local moist source density for the variables (S, Q_v, Q_c, Q_r) is

$$\mathbf{s}_{\mathrm{moist}}(Y^n) = \begin{pmatrix} h\beta_2 A(Y^n) \\ hA(Y^n) \\ -h\left[A(Y^n) + R(Y^n)\right] \\ hR(Y^n) \end{pmatrix}, \tag{25}$$

where all state-dependent quantities, including h, A, and R, are evaluated at Y^n.

The source density is mapped to the discrete finite-element tendency through the corresponding weak form. Writing M for the corresponding block finite-element mass operator and $\mathcal{B}$ for assembly of the moist-source weak form,

$$\mathbf{F}_{\mathrm{moist}} = M^{-1}\mathcal{B}(\mathbf{s}_{\mathrm{moist}}(Y^n)). \tag{26}$$

The resulting four-component discrete tendency acts on (S, Q_v, Q_c, Q_r). Embedding it into the complete DIMSWE state gives

$$\widehat{\mathbf{F}}_{\text{moist}} = \begin{pmatrix} \mathbf{0} \\ 0 \\ \mathbf{F}_{\text{moist}} \end{pmatrix}, \tag{27}$$

where $\mathbf{0}$ and 0 denote the unchanged velocity and layer-depth components. The final moist-physics update is therefore

$$X^{n+1} = Y^n + \Delta t\, \widehat{\mathbf{F}}_{\text{moist}}(Y^n). \tag{28}$$

This distinction between the local physical laws A and R, their discrete source-to-tendency mapping in (26), and the actual evaluation state Y^n is central to the learning objectives considered below. In particular, an objective that evaluates the moist physics directly at the timestep-boundary state X^n does not exactly reproduce the conditions under which that physics is deployed. Because the moist-physics update is the final child of the timestep, the prefix operator P contains no learned moist-physics evaluation. Therefore, for a truth-initialized one-step window, the state $Y^n = P(X^n)$ is independent of the learned model (in explicit sense) and may be computed and stored offline. This property will allow the one-step solver-in-the-loop objective introduced below to be written exactly as an offline deployment-consistent objective.

Because the moist-physics update is the final child of the timestep, for a truth-initialized one-step window the pre-moist-physics state $Y^n = P(X^n)$ is independent of the learned moist model and may therefore be computed and stored offline. This property will allow the one-step solver-in-the-loop objective introduced below to be written exactly as an offline deployment-consistent objective.

2.3 Learned-physics representations: network inputs, outputs, and coupling

We consider three representations for learning the local moist-physics forcings, i.e. the source terms appearing on the right-hand sides of (5)–(8) and represented discretely by (25). In all cases, the dry dynamics, hyperviscosity, moisture transport, finite-element source mapping, and timestep remain unchanged. The representations differ only in which part of the local moist source is replaced by a learned model.

2.3.1 Network inputs

For all three learned-physics representations, the neural network is evaluated using a five-component local feature vector,

$$\boldsymbol{\eta}(Z) = \begin{pmatrix} h \\ S \\ Q_v \\ Q_c \\ B \end{pmatrix}_Z, \tag{29}$$

where Z denotes the state at which the particular learning objective evaluates the moist physics.

The rain-water variable Q_r is not supplied to the network. This is consistent with the analytical moist-physics laws that are independent of Q_r. The layer depth h, however, is provided explicitly. Thus, the network inputs contain sufficient local information to construct

$$b = \frac{S}{h}, \qquad q_v = \frac{Q_v}{h}, \qquad q_c = \frac{Q_c}{h}, \tag{30}$$

and therefore to evaluate the analytical moist-physics mappings used in the present study. For the flat DoubleVortex cases considered here, $B = 0$, so the varying physical information in the feature vector is effectively (h, S, Q_v, Q_c).

2.3.2 Representation A: learned A_θ with analytical R

In the first representation, the neural network learns only the signed vapor–cloud phase-change rate,

$$A^\star(Z) \approx A_\theta(\boldsymbol{\eta}(Z))\,, \tag{31}$$

while the analytical rain-conversion law $R^\star(Z)$ given by (11) is retained. The analytical physical rate to be recovered is

$$A^\star(Z) = \min\left[\frac{q_c}{\Delta t},\ \max\left(0,\ \gamma_v \frac{q_{\text{sat}} - q_v}{\tau_v}\right)\right] - \max\left(0,\ \gamma_v \frac{q_v - q_{\text{sat}}}{\tau_v}\right). \tag{32}$$

Here $q_v = Q_v/h$ and $q_c = Q_c/h$, while q_{sat} and γ_v are themselves prescribed functions of the local state through (14) and (15). The remaining quantities entering (32) are prescribed model constants whose combined effect must be represented implicitly by the fitted network.

Thus, the physical mapping being learned in Representation A is not an intrinsically complicated unknown closure: its exact analytical form is known and consists primarily of state-dependent algebraic operations, thresholding, and prescribed scalar parameters. The purpose of the study is therefore not to discover an unknown physical law, but to examine how different learning objectives recover and deploy a known law within the numerical model.

The corresponding local moist source is explicitly assembled from the network output,

$$\mathbf{s}_{A,\theta}(Z) = h \begin{pmatrix} \beta_2 A_\theta(\boldsymbol{\eta}(Z)) \\ A_\theta(\boldsymbol{\eta}(Z)) \\ -\left[A_\theta(\boldsymbol{\eta}(Z)) + R^\star(Z)\right] \\ R^\star(Z) \end{pmatrix}. \tag{33}$$

Thus the learned model replaces only one physical rate while retaining the analytical rain law and the prescribed moist-source structure.

2.3.3 Representation B: learned (A_θ, R_θ)

In the second representation, both scalar moist-process rates are learned according to,

$$(A_\theta(\boldsymbol{\eta}(Z)), R_\theta(\boldsymbol{\eta}(Z))) = \mathcal{N}_\theta(\boldsymbol{\eta}(Z))\,, \tag{34}$$

In addition to recovering the phase-change law in (32), the network is therefore asked to reproduce the analytical cloud-to-rain conversion law

$$R^\star(Z) = \max\left[0,\ \gamma_r \frac{q_c - q_{\text{precip}}}{\tau_r}\right]. \tag{35}$$

This is again a relatively simple algebraic mapping: $q_c = Q_c/h$ is state-dependent, while q_{precip}, γ_r, and τ_r are prescribed model parameters. The learned model is nevertheless not explicitly informed of the threshold or non-negativity structure in (35).

The two network outputs are then assembled manually through the prescribed moist-source structure,

$$\mathbf{s}_{B,\theta}(Z) = h \begin{pmatrix} \beta_2 A_\theta(\boldsymbol{\eta}(Z)) \\ A_\theta(\boldsymbol{\eta}(Z)) \\ -\left[A_\theta(\boldsymbol{\eta}(Z)) + R_\theta(\boldsymbol{\eta}(Z))\right] \\ R_\theta(\boldsymbol{\eta}(Z)) \end{pmatrix}. \tag{36}$$

Thus, even when the learned rates themselves are inaccurate, the resulting source vector continues to satisfy the prescribed total-water and thermodynamic identities. However, the unconstrained learned rate R_θ is not guaranteed to recover the non-negativity or threshold activation of the analytical rain law. Because the true mapping is simple and known, departures from these properties provide a particularly useful diagnostic of whether accurate hybrid-state evolution also implies recovery of the underlying physical process.

2.3.4 Representation C: learned moist-source vector

The third representation removes the explicit A–R decomposition and learns the four moist-source components directly,

$$\mathbf{s}_{C,\theta}(\boldsymbol{\eta}(Z)) = \mathcal{N}_\theta(\boldsymbol{\eta}(Z)) = \begin{pmatrix} T_{S,\theta} \\ T_{v,\theta} \\ T_{c,\theta} \\ T_{r,\theta} \end{pmatrix}. \tag{37}$$

Unlike Representations A and B, no algebraic constraint forces these four learned components to satisfy the source relations in (25), total-water conservation, or the prescribed thermodynamic coupling. Representation C therefore tests whether these relationships are recovered from training alone rather than imposed through the model representation. The target mappings in Representation C are nevertheless not intrinsically complicated. Relative to the scalar rates learned in Representations A and B, the four target source components are obtained through simple state-dependent multiplications and linear combinations,

$$T_S^\star = h\beta_2 A^\star, \qquad T_v^\star = hA^\star, \qquad T_c^\star = -h(A^\star + R^\star), \qquad T_r^\star = hR^\star. \tag{38}$$

Thus Representation C does not present the network with a fundamentally more complicated unknown physical law; rather, it removes the known low-dimensional structure relating the four tendencies.

The analytical source vector is completely determined by only two scalar rates, A and R, since

$$\mathbf{s}^\star_{\text{moist}} = hA^\star \begin{pmatrix} \beta_2 \\ 1 \\ -1 \\ 0 \end{pmatrix} + hR^\star \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}. \tag{39}$$

Thus, although the moist source has four components, the physically admissible source vectors occupy only a two-dimensional subspace of this four-dimensional output space, since they are completely determined by the two scalar rates A and R. Representations A and B enforce this low-dimensional structure by construction, whereas Representation C predicts the four source components independently. Therefore, Representation C has sufficient freedom to produce source tendencies that cannot be associated with any pair of physical rates (A, R) and that may violate the total-water and thermodynamic identities derived above. The comparison therefore tests whether the training objective alone is sufficient to recover this known physical structure when it is not imposed explicitly.

2.3.5 Notes on improving training later

Although the current feature vector contains sufficient information to evaluate the analytical moist-physics laws, the network is supplied with the conservative variables (h, S, Q_v, Q_c, B) rather than the specific variables in which those laws are naturally expressed. It must therefore represent the transformations

$$b = \frac{S}{h}, \qquad q_v = \frac{Q_v}{h}, \qquad q_c = \frac{Q_c}{h} \tag{40}$$

internally before approximating the analytical phase-change and rain laws.

A useful follow-on comparison is therefore between the present feature map

$$(h, S, Q_v, Q_c, B) \tag{41}$$

and the physically transformed feature map

$$(h, b, q_v, q_c, B). \tag{42}$$

These feature sets contain equivalent local physical information for the present parameterization, but their numerical conditioning and ease of function approximation may differ. For the flat DoubleVortex cases, $B = 0$ in both representations.

2.4 Machine-learning objectives: training loss functions

We consider four training objectives that progressively incorporate more of the numerical evolution surrounding the learned moist-physics parameterization. Denoting the trusted timestep-boundary trajectory by $X_n^\star$ and the state immediately before the moist-physics update by

$$Y_n^\star = P(X_n^\star), \tag{43}$$

the objectives are:

1. M1-Y: direct regression of the analytical moist-physics law at $Y_n^\star$;
2. M2-Y/H1: one-step training against the discrete effect of the moist physics at $Y_n^\star$;
3. H2: two-step recursive training; and
4. H5: five-step recursive training.

Because the moist-physics update is the final component of the split DIMSWE timestep, $Y_n^\star = P(X_n^\star)$ can be computed independently of the learned model. Consequently, both M1-Y and M2-Y/H1 can be formulated entirely from stored truth-derived states. H2 is the first objective for which a state modified by an earlier learned-physics prediction is subsequently supplied to the network, while H5 extends this recursive feedback over five timesteps.

2.4.1 M1-Y: direct physical-law regression

M1-Y trains the network directly against the known analytical moist physics at the state where the parameterization is evaluated. Let $\mathbf{z}^\star(Z)$ denote the analytical quantity corresponding to the selected learned-physics representation,

$$\mathbf{z}^\star(Z) = \begin{cases} A^\star(Z), & \text{Representation A,} \\ (A^\star(Z), R^\star(Z))^T, & \text{Representation B,} \\ \mathbf{s}^\star_{\mathrm{moist}}(Z), & \text{Representation C.} \end{cases} \tag{44}$$

The direct-regression loss is

$$\mathcal{L}_{\mathrm{M1-Y}}(\theta) = \frac{1}{N}\sum_{n=1}^{N} \|\mathbf{z}_\theta(\boldsymbol{\eta}(Y_n^\star)) - \mathbf{z}^\star(Y_n^\star)\|^2 . \tag{45}$$

Thus M1-Y is a conventional offline regression of the local physical law, evaluated at the same pre-moist state on which the network acts during the numerical timestep.

2.4.2 M2-Y / H1: one-step discrete training

M2-Y/H1 retains the same pre-moist truth state $Y_n^\star$ but changes the quantity being fitted. Because the moist Euler update is the final component of the timestep, the discrete moist tendency required to reproduce the trusted one-step transition is

$$\widehat{\mathbf{F}}^{Y}_{n,\mathrm{target}} = \frac{X^\star_{n+1} - Y^\star_n}{\Delta t}. \tag{46}$$

The corresponding tendency-based objective is

$$\mathcal{L}_{\mathrm{M2-Y}}(\theta) = \frac{1}{N}\sum_{n=1}^{N} \left\| \widehat{\mathbf{F}}_{\mathrm{moist},\theta}(Y_n^\star) - \frac{X^\star_{n+1} - Y^\star_n}{\Delta t} \right\|^2 . \tag{47}$$

The same training problem can be written as a one-step state-prediction objective. Starting from $X_n^\star$, the timestep prefix produces $Y_n^\star$, after which the learned moist update gives

$$\widehat{X}_{n+1} = Y_n^\star + \Delta t\, \widehat{\mathbf{F}}_{\mathrm{moist},\theta}(Y_n^\star). \tag{48}$$

The H1 loss is therefore

$$\mathcal{L}_{\mathrm{H1}}(\theta) = \frac{1}{N}\sum_{n=1}^{N} \left\| Y_n^\star + \Delta t\, \widehat{\mathbf{F}}_{\mathrm{moist},\theta}(Y_n^\star) - X^\star_{n+1} \right\|^2 . \tag{49}$$

Using (46), the two losses satisfy

$$\mathcal{L}_{\mathrm{H1}} = \Delta t^2\, \mathcal{L}_{\mathrm{M2-Y}}, \tag{50}$$

provided the same fixed state-component scaling is used. M2-Y and H1 therefore define the same optimization problem up to a constant factor.

H1 contains no recursive model-state dependence. Since $Y_n^\star = P(X_n^\star)$ is independent of the network parameters,

$$\frac{\partial \widehat{X}_{n+1}}{\partial \theta} = \Delta t \, \frac{\partial \widehat{\mathbf{F}}_{\text{moist},\theta}(Y_n^\star)}{\partial \theta}. \tag{51}$$

The one-step problem can therefore be constructed entirely offline from stored $Y_n^\star$ and $X_{n+1}^\star$. We use M2-Y to denote this offline tendency formulation and H1 to denote the equivalent one-step state formulation.

2.4.3 H2 and H5: recursive rollout objectives

To define the recursive objectives, write the complete learned DIMSWE timestep as

$$\mathcal{F}_\theta(X) = P(X) + \Delta t \, \widehat{\mathbf{F}}_{\text{moist},\theta}(P(X)) \,. \tag{52}$$

For a training window beginning from a trusted state,

$$\widehat{X}_{n,0} = X_n^\star, \qquad \widehat{X}_{n,j+1} = \mathcal{F}_\theta(\widehat{X}_{n,j}), \qquad j = 0, \ldots, H-1. \tag{53}$$

The accumulated rollout objective is

$$\mathcal{L}_H(\theta) = \frac{1}{N_H} \sum_n \sum_{j=1}^{H} \left\| \widehat{X}_{n,j} - X_{n+j}^\star \right\|^2, \tag{54}$$

where fixed normalization factors used in the implemented loss are suppressed for clarity, with

$$H = 2 \quad \text{for H2}, \qquad H = 5 \quad \text{for H5}. \tag{55}$$

For the first step of every training window,

$$P(\widehat{X}_{n,0}) = P(X_n^\star) = Y_n^\star, \tag{56}$$

so the first network evaluation is performed on a fixed truth-derived state. After the first learned moist update, however, $\widehat{X}_{n,1}$ depends on θ. Subsequent parameter sensitivities therefore satisfy

$$\frac{d\widehat{X}_{n,j+1}}{d\theta} = \frac{\partial \mathcal{F}_\theta}{\partial X}(\widehat{X}_{n,j}) \frac{d\widehat{X}_{n,j}}{d\theta} + \frac{\partial \mathcal{F}_\theta}{\partial \theta}(\widehat{X}_{n,j}), \tag{57}$$

with

$$\frac{d\widehat{X}_{n,0}}{d\theta} = 0. \tag{58}$$

Thus H2 is the first objective for which the network is trained on a state that has already been modified by an earlier learned-physics prediction. H5 applies the same construction over five successive timesteps and therefore exposes the network to a longer sequence of model-generated states.

Training objective	Information included in the loss
M1-Y	local analytical moist physics at $Y^\star$
M2-Y / H1	discrete one-step state effect
H2	two-step recursive model-state feedback
H5	five-step recursive model-state feedback

The progression from M1-Y to M2-Y/H1 therefore isolates the additional effect of fitting the discrete one-step state response rather than the local analytical physical law. H2 and H5 then introduce the qualitatively new ingredient of recursive model-state feedback over progressively longer rollouts.

2.5 Neural-network architecture and training details

All learned-physics representations use local fully connected neural networks that act independently at each deployed spatial point. The common raw input vector is

$$\boldsymbol{\eta} = (h, S, Q_v, Q_c, B)^T\,, \tag{59}$$

as introduced above. Prior to evaluation by the network, each input component is normalized as

$$\widetilde{\eta}_k = \frac{\eta_k - \mu_k}{\sigma_k}, \tag{60}$$

where the offsets μ_k and scales σ_k are computed once from the ground-truth states used for training using carrier-mass-weighted population statistics. These values are then held fixed throughout training and deployment. During deployment, the evolving DIMSWE state is normalized using these same constants before being supplied to the neural network. For the constant topography input $B = 0$, the offset is zero and the scale is set to unity, so this component is unchanged by normalization.

The neural network has two fully connected hidden layers with hyperbolic tangent activations and a linear output layer,

$$\mathcal{N}_\theta(\widetilde{\boldsymbol{\eta}}) = W_3 \tanh[W_2 \tanh(W_1 \widetilde{\boldsymbol{\eta}} + \mathbf{b}_1) + \mathbf{b}_2] + \mathbf{b}_3. \tag{61}$$

The architectures are

Representation	Network architecture	Number of parameters
A	5−32−32−1	1281
B	5−32−32−2	1314
C	5−32−32−4	1380

The output dimension therefore changes with the learned-physics representation, while the input and hidden-layer architecture are held fixed. All networks use double precision. The three representations consequently have similar total parameter counts despite predicting different numbers of physical quantities. This common architecture provides a simple controlled comparison, but does not guarantee that the representations are matched in effective expressive capacity. Sensitivity to network width and capacity is left for future study.

Initial weights are generated using a seed-zero Glorot-uniform initialization independently for each layer, with all biases initialized to zero.

The learned outputs are also scaled when evaluating the optimization objectives. No output centering is applied. For a learned quantity z_j, the corresponding normalized residual is

$$r_j = \frac{z_{\theta,j} - z_j^\star}{\sigma_{z_j}}. \tag{62}$$

Representation A uses

$$\sigma_A = 9.0522 \times 10^{-8}. \tag{63}$$

Representation B uses the same scale for A and

$$\sigma_R = 1.9903 \times 10^{-11} \tag{64}$$

for the rain rate. The latter is computed from the rain-active portion of the training data, although samples for which $R^\star = 0$ remain in the training loss. Representation C uses separate scales for its four source components, in the ordering (S, Q_v, Q_c, Q_r),

$$\boldsymbol{\sigma}_C = \begin{pmatrix} 6.6715 \times 10^{-3} \\ 6.8034 \times 10^{-5} \\ 6.8034 \times 10^{-5} \\ 1.5076 \times 10^{-8} \end{pmatrix}. \tag{65}$$

Optimization is performed through the ROL/PyROL interface using a line-search limited-memory BFGS method with a maximum secant storage of 20. Parameter derivatives are supplied through the JAX-based learned-physics implementation together with the discrete differentiation required by the corresponding objective. Hessian-vector-product machinery was also developed and verified but is not used in the optimizations considered here.

M1-Y is initialized directly from the common seed-zero network and is trained for a maximum of 10,000 iterations. The H1 calculation was performed before M1-Y was introduced and was initialized from the direct-regression model available at that stage of the study. H2 is initialized from the corresponding H1 model, and H5 is subsequently initialized from H2. Consequently, H1, H2, and H5 form a sequential optimization sequence, while M1-Y is an independently initialized direct-regression fit.

The prescribed optimization budgets are

Training objective	Initialization	Maximum iteration budget
M1-Y	seed zero	10000
H1 / M2-Y	previous direct-regression model	5000
H2	H1	20
H5	H2	20

Thus comparisons among M1-Y, H1, H2, and H5 should account for both the different training objectives and their different optimization histories. In particular, H2 and H5 are short continuation stages rather than independently initialized fits.

3 Ground-Truth DoubleVortex Simulations

The learning studies are based on two trusted DoubleVortex simulations in which the moist-physics parameterization is evaluated analytically rather than replaced by a neural network. These simulations provide both the trajectories used for training and the reference solutions against which the learned models are evaluated. The two cases share the same underlying vortex construction and physical parameters, but differ in spatial resolution and initial moisture state. Test 2A provides a relatively simple non-raining regime, whereas Test 2B is designed to evolve through condensation, cloud-water production, and subsequent rain formation (Test 1 was the hyperviscosity exercise).

3.1 Initial and boundary conditions

Both DoubleVortex cases are posed on a flat, doubly periodic square domain,

$$\Omega = [0, L_x) \times [0, L_y), \qquad L_x = L_y = L = 5000 \text{ km}, \tag{66}$$

with zero bottom topography,

$$B = 0. \tag{67}$$

The reference layer depth is

$$H_0 = 750 \text{ m}, \tag{68}$$

and the depth-perturbation scale is

$$\Delta h = 75 \text{ m}. \tag{69}$$

The two vortex centers are located at

$$(x_1, y_1) = (2000, 2000) \text{ km}, \qquad (x_2, y_2) = (3000, 3000) \text{ km}, \tag{70}$$

and the Gaussian width parameters are

$$\sigma_x = \sigma_y = 375 \text{ km}. \tag{71}$$

The simulations use

$$g = 9.80616 \text{ m}\,\text{s}^{-2}, \qquad f = 6.147 \times 10^{-5} \text{ s}^{-1}, \tag{72}$$

and a timestep

$$\Delta t = 100 \text{ s}. \tag{73}$$

Each ground-truth trajectory is integrated for 160 timesteps to $t_f = 16000$ s, with the state stored every timestep, giving 161 stored states including the initial condition.

3.1.1 DoubleVortex construction and geostrophic flow

The two vortices are constructed using a smooth periodicization of the horizontal coordinates. For vortex $i = 1, 2$, define

$$X_i(x) = \frac{L_x}{\pi\sigma_x} \sin\left(\frac{\pi(x - x_i)}{L_x}\right), \qquad Y_i(y) = \frac{L_y}{\pi\sigma_y} \sin\left(\frac{\pi(y - y_i)}{L_y}\right), \tag{74}$$

with

$$\sigma_x = \sigma_y = \frac{3}{40} L_x = 375 \text{ km}. \tag{75}$$

The corresponding periodicized Gaussian profiles are

$$G_i(x, y) = \exp\left[-\frac{1}{2}\left(X_i^2 + Y_i^2\right)\right]. \tag{76}$$

Near each vortex center this reduces locally to the usual Gaussian form $\exp[-r^2/(2\sigma^2)]$.

The initial layer depth is

$$h_0(x, y) = H_0 - \Delta h \left[G_1(x, y) + G_2(x, y) - \frac{4\pi\sigma_x\sigma_y}{L_x L_y}\right], \tag{77}$$

so that the two vortices correspond to negative layer-depth anomalies. The final constant offset is part of the implemented DoubleVortex construction and shifts the background depth without affecting the analytical velocity, since its spatial gradient vanishes.

To write the analytical derivatives compactly, define

$$\widetilde{X}_i(x) = \frac{L_x}{2\pi\sigma_x} \sin\left(\frac{2\pi(x - x_i)}{L_x}\right), \qquad \widetilde{Y}_i(y) = \frac{L_y}{2\pi\sigma_y} \sin\left(\frac{2\pi(y - y_i)}{L_y}\right). \tag{78}$$

Then

$$\frac{\partial h_0}{\partial x} = \frac{\Delta h}{\sigma_x} \sum_{i=1}^{2} \widetilde{X}_i G_i, \qquad \frac{\partial h_0}{\partial y} = \frac{\Delta h}{\sigma_y} \sum_{i=1}^{2} \widetilde{Y}_i G_i. \tag{79}$$

The initial velocity is constructed directly from these analytical derivatives,

$$u_0 = -\frac{g}{f}\frac{\partial h_0}{\partial y}, \qquad v_0 = \frac{g}{f}\frac{\partial h_0}{\partial x}. \tag{80}$$

Using the rotation operator

$$\mathcal{R}(\mathbf{v}) = (-v, u)^T, \tag{81}$$

it follows immediately that

$$f\mathcal{R}(\mathbf{v}_0) = -g\nabla h_0, \tag{82}$$

and hence

$$f\mathcal{R}(\mathbf{v}_0) + g\nabla h_0 = 0. \tag{83}$$

Thus the initial velocity and layer-depth perturbation are constructed to be in exact geostrophic balance with respect to the constant-g height-gradient and Coriolis terms.

The specific thermal variable $b = S/h$ is initialized independently as

$$b_0(x, y) = g\left[1 + c\exp\left(-\frac{(x - L_x/2)^2 + (y - L_y/2)^2}{a^2 D^2}\right)\right], \tag{84}$$

where

$$c = 0.05, \qquad a = \frac{1}{3}, \qquad D = \frac{L_x}{2}, \tag{85}$$

and therefore

$$S_0(x, y) = h_0(x, y) b_0(x, y). \tag{86}$$

Unlike the layer-depth vortices, this thermal perturbation is constructed directly from Cartesian distance to the center of the domain rather than from the sine-periodicized coordinates used in Eq. (76).

With $B = 0$, the initial saturation mixing ratio is

$$q_{\text{sat},0} = q_0 \frac{H_0}{h_0} \exp\left[20\left(1 - \frac{b_0}{g}\right)\right], \qquad q_0 = 0.002, \tag{87}$$

and the initial moisture fields are

$$Q_{v,0} = h_0(1 - \zeta) q_{\text{sat},0}, \qquad Q_{c,0} = Q_{r,0} = 0. \tag{88}$$

Although Eq. (83) is exact for the Coriolis–constant-g height-gradient pair, the complete moist thermal shallow-water initial condition is not asserted to be a steady solution of the full equations. The nonlinear and thermal terms, together with transport, dissipation, and moist-physics processes, remain active during the subsequent evolution.

The resulting common DoubleVortex initial state is shown in Fig. 1. The negative layer-depth anomalies are associated with two compact positive-vorticity cores and circulating geostrophic flow, while the thermal/buoyancy perturbation is broader and centered between the two vortices. This dynamical and thermal structure is common to Tests 2A and 2B; their different moisture initializations are introduced below.

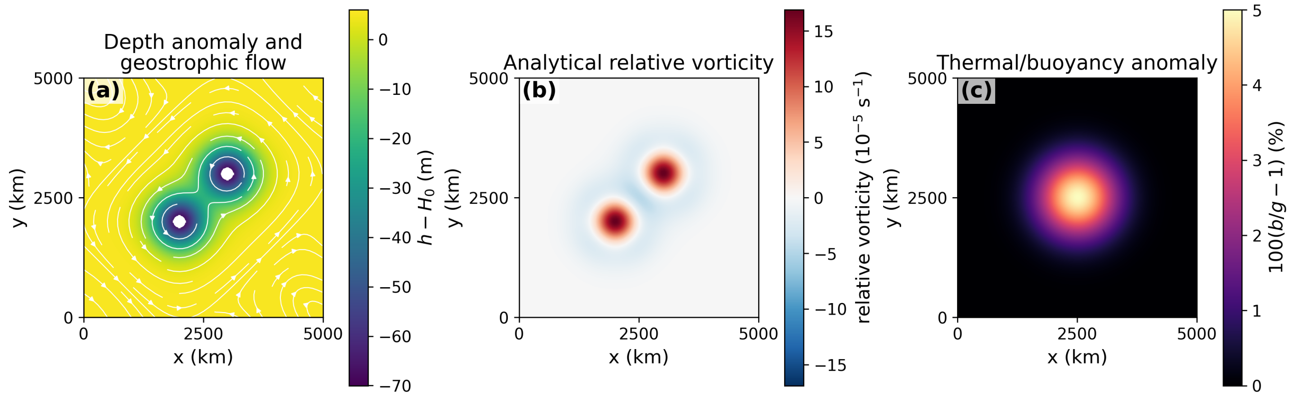

Figure 1: Common DoubleVortex initial condition. (a) Layer-depth anomaly $h - H_0$ with the analytically constructed geostrophic flow. (b) Corresponding analytical relative vorticity. (c) Central thermal/buoyancy anomaly $100(b/g - 1)$, where $b = S/h$.

3.2 Test 2A: controlled non-raining case

Test 2A uses a 16×16 horizontal mesh and initializes the vapor field at saturation,

$$\zeta = 0, \qquad q_v(\mathbf{x}, 0) = q_{\text{sat}}(\mathbf{x}, 0). \tag{89}$$

The case provides a relatively inexpensive controlled setting in which condensation and evaporation are active, while rain production does not become dynamically important. It therefore provides a useful first test of learned phase-change physics without the additional complication of an active rain-production rate.

3.3 Test 2B: rain-active case

Test 2B increases the horizontal resolution to 64×64 cells and modifies the initial vapor field to produce a modest supersaturation,

$$\zeta = -0.06, \tag{90}$$

so that

$$q_v(\mathbf{x}, 0) = 1.06\, q_{\text{sat}}(\mathbf{x}, 0). \tag{91}$$

All other principal DoubleVortex parameters are retained from Test 2A. The enhanced initial moisture content drives stronger condensation and cloud-water production, eventually activating the rain-conversion rate R. Test 2B therefore provides the more complete moist-physics problem, in which both the net phase-change rate $A = E - C$ and the rain-production rate R are dynamically relevant. It is therefore the principal case used for the comparison of Representations A, B, and C.

3.4 Ground-truth evolution and physical diagnostics

Having established the common DoubleVortex flow and the distinct moisture initializations of Tests 2A and 2B, we now examine how these differences affect the subsequent moist evolution. Test 2A remains a non-raining phase-change case, whereas Test 2B undergoes rapid condensation, cloud-water accumulation, and eventual activation of the rain-production process. The temporal evolution of both regimes is summarized in Fig. 2, followed by the spatial development of the rain-active Test 2B case and the evolution of its underlying vortex structure in Fig. 3.

Test 2A remains non-raining throughout the calculation. Its maximum cloud-water mixing ratio is 3.12×10^{-5} kg kg^{-1}, approximately 31% of the precipitation threshold $q_{\text{precip}} = 10^{-4}$ kg kg^{-1}, and therefore $R = Q_r = 0$ throughout. Test 2A nevertheless develops regions of both condensation and evaporation as the vortex flow reorganizes the moisture field, making it a useful controlled phase-change case without active cloud-to-rain conversion.

Test 2B evolves very differently. The imposed 6% supersaturation is consumed rapidly because the condensation timescale is equal to the numerical timestep, $\tau_v = \Delta t = 100$ s. The initial moist update therefore converts a large fraction of the excess vapor directly into cloud water, producing the nearly instantaneous increase in integrated cloud water visible in Fig. 2(a). After this initial condensation pulse, the vortex flow redistributes the cloud and vapor fields and produces coexisting regions of condensation and evaporation.

The maximum local cloud water eventually reaches the precipitation threshold. The rain production rate R first becomes nonzero at

$$t = 5100 \text{ s} \approx 1.42 \text{ h}, \tag{92}$$

and remains active thereafter (Fig. 2(b)). A sustained-rain interval is certified by $t = 6100$ s. The largest local cloud-water concentration and local rain-production rate occur near $t = 8900$ s, whereas the domain-integrated rain-production rate reaches its maximum later, at

$$t = 12000 \text{ s} \approx 3.33 \text{ h}. \tag{93}$$

The distinction between these times emphasizes that the first rain event is a local threshold crossing, whereas the later maximum in integrated rain production reflects the spatial growth of the rain-producing region.

The corresponding spatial evolution of Test 2B is shown in Fig. 3. Prior to rain onset, cloud water is concentrated around the two vortex regions while R remains zero. Once the cloud-water threshold is crossed, rain production first appears in narrow localized arcs. These regions are then stretched and wrapped by the vortex flow into extended curved bands. By the mature-rain state at $t = 12000$ s, substantial rain production occurs along two broad spiral-like structures surrounding the vortex cores, while accumulated rain water occupies a larger region downstream of the actively raining bands. The final state retains this wrapped two-lobed structure even though the strongest local rain-production rate has already decreased from its earlier maximum. This persistence is expected because the present moist-physics model contains no rain-removal process such as fallout or rain re-evaporation; Q_r is therefore transported within the periodic domain once produced.

The strong deformation visible in Fig. 3 and in the truth-evolution movie occurs with remarkably little translation of the underlying vortex pair. Figure 4 tracks the two positive-vorticity cores

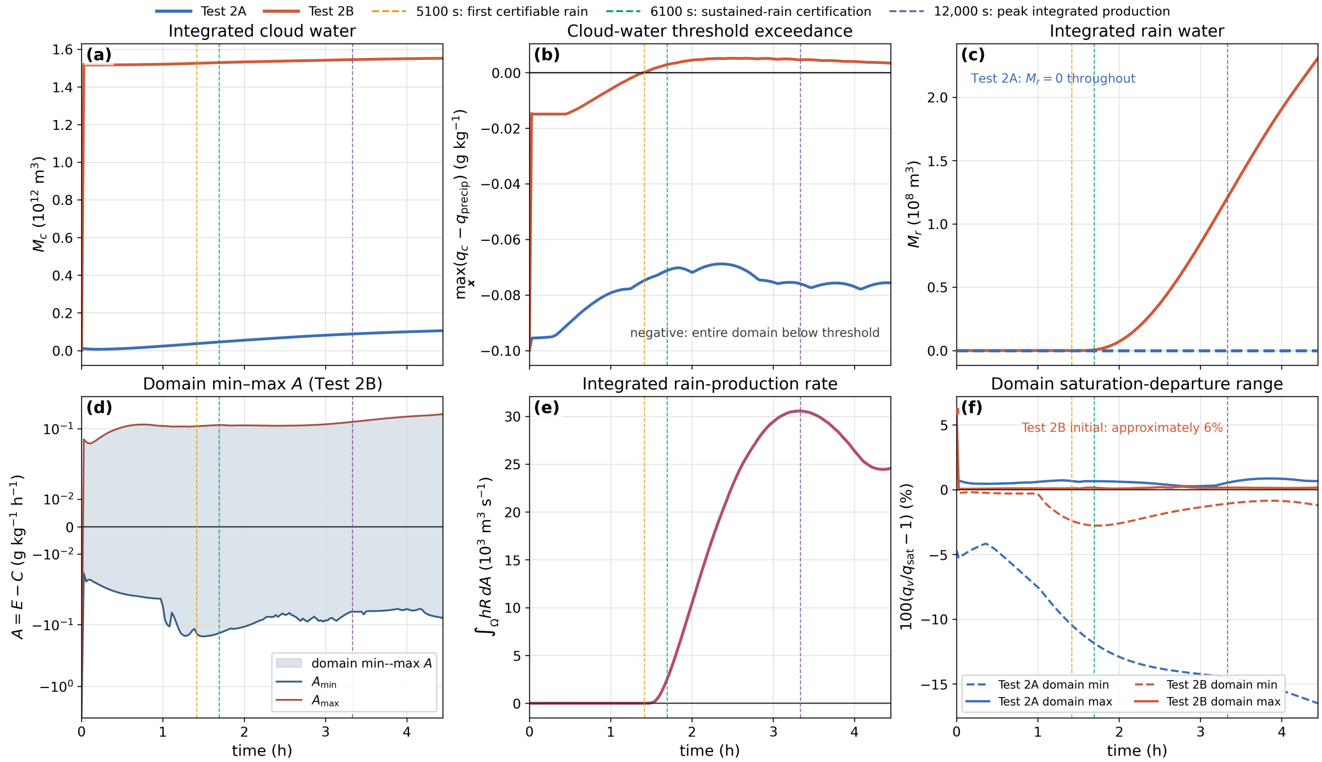

Figure 2: Ground-truth temporal diagnostics for Test 2A and Test 2B, with emphasis on the Test 2B transition to rain. (a) Integrated cloud-water reservoir M_c. (b) Maximum cloud-water threshold exceedance $\max_{\mathbf{x}}(q_c - q_{\mathrm{precip}})$; negative values indicate that the entire domain remains below the rain threshold. (c) Integrated rain-water reservoir M_r. (d) Domain minimum and maximum of the net phase-change rate $A = E - C$ for Test 2B. (e) Domain-integrated Test 2B rain-production rate $\int_\Omega hR\,dA$. (f) Domain minimum and maximum of the post-prefix saturation departure $100(q_v/q_{\mathrm{sat}} - 1)$ for both cases. Vertical dashed lines mark first certifiable rain at 5100 s, sustained-rain certification at 6100 s, and peak integrated rain production at 12000 s. Test 2A and Test 2B differ in both initial vapor loading and spatial resolution, so their comparison characterizes distinct moist-physics regimes rather than constituting a grid-convergence study.

using continuously followed local vorticity-weighted centroids. Over the full 16000 s simulation, the maximum displacement of either core is only 18.0 km, or 0.36% of the domain size and 4.8% of the characteristic vortex width $\sigma = 375$ km. The pair separation changes by approximately 27 km and its orientation by less than 1°, while the pair midpoint remains fixed to numerical precision. Thus the dominant moist evolution is material deformation and wrapping around an approximately stationary vortex skeleton rather than bulk translation of the vortices.

4 Machine-Learning Results

4.1 Training and evaluation data

The machine-learning results presented below use the higher-resolution 64×64 rain-active DoubleVortex trajectory described in Sec. 3. The trajectory contains 161 stored states, indexed from 0 to 160. States 0–80 are used for training, while states 81–160 are reserved for evaluation after training. The latter therefore assess temporal generalization beyond the portion of the trajectory used for fitting.

Each stored state contributes 65,536 local spatial samples (64^2 cells $\times 4^2$ Gauss–Lobatto–

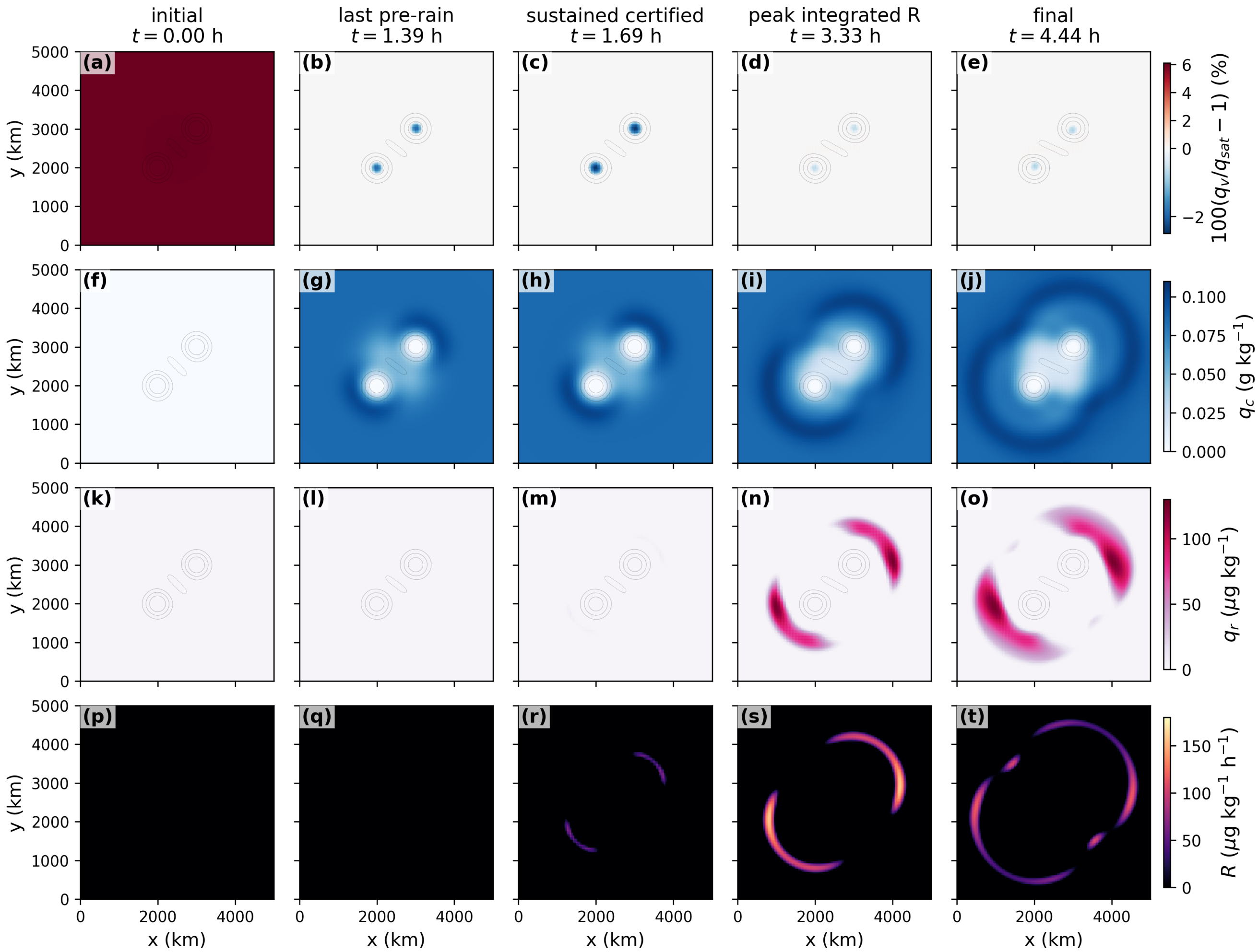

Figure 3: Spatial evolution of the Test 2B ground-truth solution at five physically selected times: the initial state, the last pre-rain state, the sustained-rain state, the state of peak integrated rain production, and the final state. Rows show the saturation departure $100(q_v/q_{\text{sat}} - 1)$, specific cloud water q_c, specific rain water q_r, and the rain-production rate R, respectively. Gray relative-vorticity contours indicate the locations of the two vortex cores. Rain production first develops in narrow arcs near regions of enhanced cloud water and is subsequently stretched into extended curved bands by the vortex flow.

Legendre quadrature points per cell), giving 5,308,416 ($65{,}536 \times 81$) training samples across states 0–80 and 5,242,880 evaluation samples across states 81–160. Not all of them are active rain samples.

4.2 Optimization of the training objectives

Figure 5 shows the reduction of the M1-Y, H1, H2, and H5 training objectives for the three learned-physics representations. Because network parameters were retained only at selected iterations, the curves show the objective at those saved points rather than at every optimization iteration.

For all three representations, M1-Y reduces the direct physical-law objective by several orders of magnitude over the 10,000 iterations. In this initial exploratory investigation we set a maximum of 10,000 for $M1 - Y$ unless the objective falls below 10^{-8}. Thus the we have not fully optimized the Network parmeters yet.

The H1/M2-Y objective also decreases throughout its 5,000-iteration optimization for all three

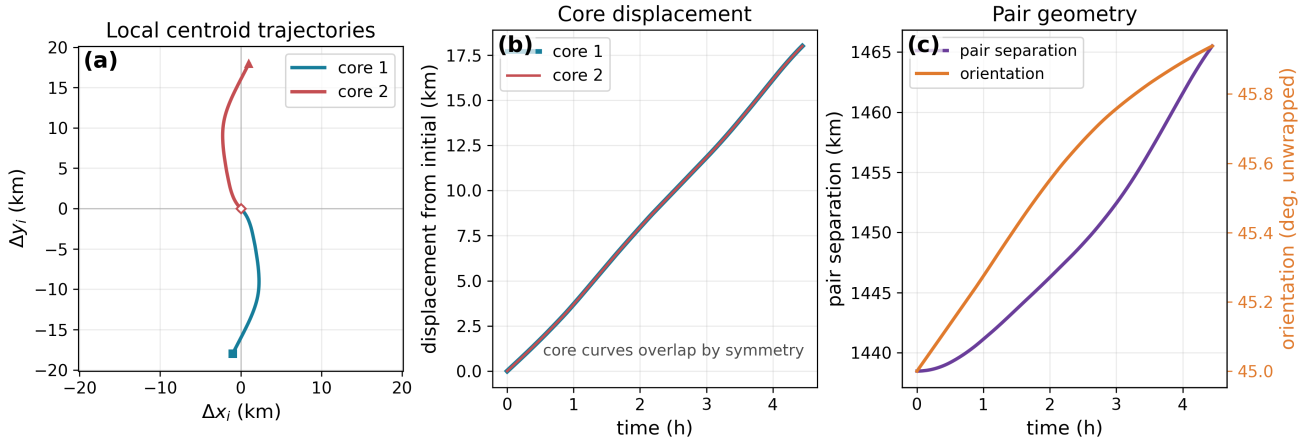

Figure 4: Motion of the two positive-vorticity cores during the Test 2B ground-truth simulation, measured using continuously tracked local vorticity-weighted centroids. (a) Core trajectories relative to their respective initial positions, shown as $(\Delta x_i, \Delta y_i)$. (b) Displacement magnitude of each core from its initial position; the two curves nearly overlap because of the symmetry of the vortex pair. (c) Separation and unwrapped orientation of the vortex pair. The maximum core displacement is only about 18 km, showing that the pronounced evolution of the cloud and rain fields is dominated by deformation and advection around a nearly stationary vortex pair.

representations. This is also stopped by a cutoff and hence H1 network could also be further optimized. H1 begins from an already trained direct-regression model, and its one-step objective is therefore already relatively small at the start of this optimization stage.

H2 is initialized from H1, and H5 is subsequently initialized from H2. Each recursive stage is limited to 20 iterations. Only their initial and final objective values are shown. These short continuation stages produce little change for Representations A and B, whereas Representation C shows a more noticeable reduction (it starts from a higher objective value than A and B). The H2 and H5 results should therefore be interpreted together with their short optimization budgets and sequential initialization.

The numerical values of M1-Y, H1, H2, and H5 are not directly comparable because the objectives measure different quantities. Figure 5 is therefore used only to assess the reduction achieved for each objective during its own optimization.

4.3 Training and evaluation errors

We next examine whether the quantities minimized during training remain accurate over the later portion of the DoubleVortex trajectory. Figure 6 compares each fitted objective over the training states 0–80 (solid curves) and the evaluation states 81–160 (dashed curves).

For M1-Y, the network is trained directly against the local analytical moist physics at the pre-moist states $Y^\star$. The training objective decreases by several orders of magnitude for all three representations. The corresponding evaluation error remains substantially larger, however, indicating that the accuracy obtained over the fitted portion of the trajectory does not extend uniformly to its later evolution. The M1-Y training objective is still decreasing at the end of the prescribed 10,000 iterations, so additional optimization could further improve the fit to the training data. Whether it would also reduce the evaluation error is not established by the present calculations.

A similar training–evaluation separation is observed for H1/M2-Y, but here the quantity being minimized is the one-step state error rather than the local moist-physics error. The state error

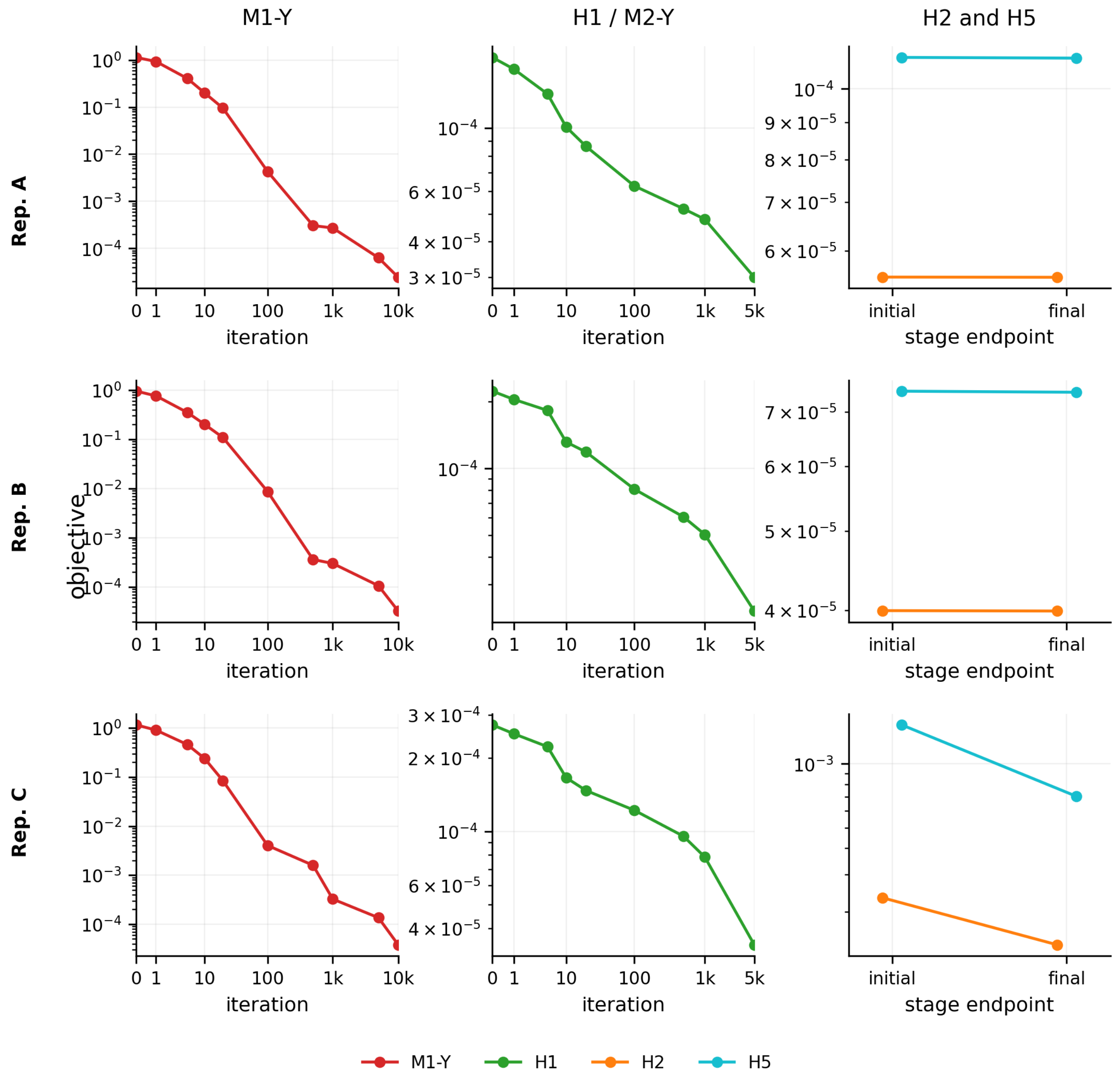

Figure 5: Optimization of the Test 2B training objectives for Representations A–C. The left column shows M1-Y, which fits the local analytical moist physics at the pre-moist state $Y^\star$. The middle column shows the one-step H1/M2-Y objective. The right column shows the initial and final values of the recursive H2 and H5 objectives. Markers in the M1-Y and H1 panels indicate iterations at which the network parameters were saved. H2 and H5 were each limited to 20 iterations, and only their initial and final objective values are shown. Because the objectives and their normalizations differ, numerical values should not be compared between different objective types or representations.

becomes small over the training interval, while the same one-step prediction problem remains considerably more difficult over the evaluation interval.

The distinction between the two objectives is worth noting. In M1-Y the loss acts directly on the quantities predicted by the network: A, (A, R), or the moist-source vector. In H1/M2-Y, the network outputs first pass through the discrete source-to-state mapping before the loss is evaluated on the resulting state increment. The M1-Y regression problem is therefore more directly tied to the network outputs, whereas H1 asks the optimizer to infer the local physics through its effect on the numerical state. The results in Fig. 6 suggest that the simpler direct regression is easier to fit and generalize in the present configuration. This conclusion is specific to the network size, training data, and optimization budgets used here; greater model capacity or broader training data could alter the comparison.

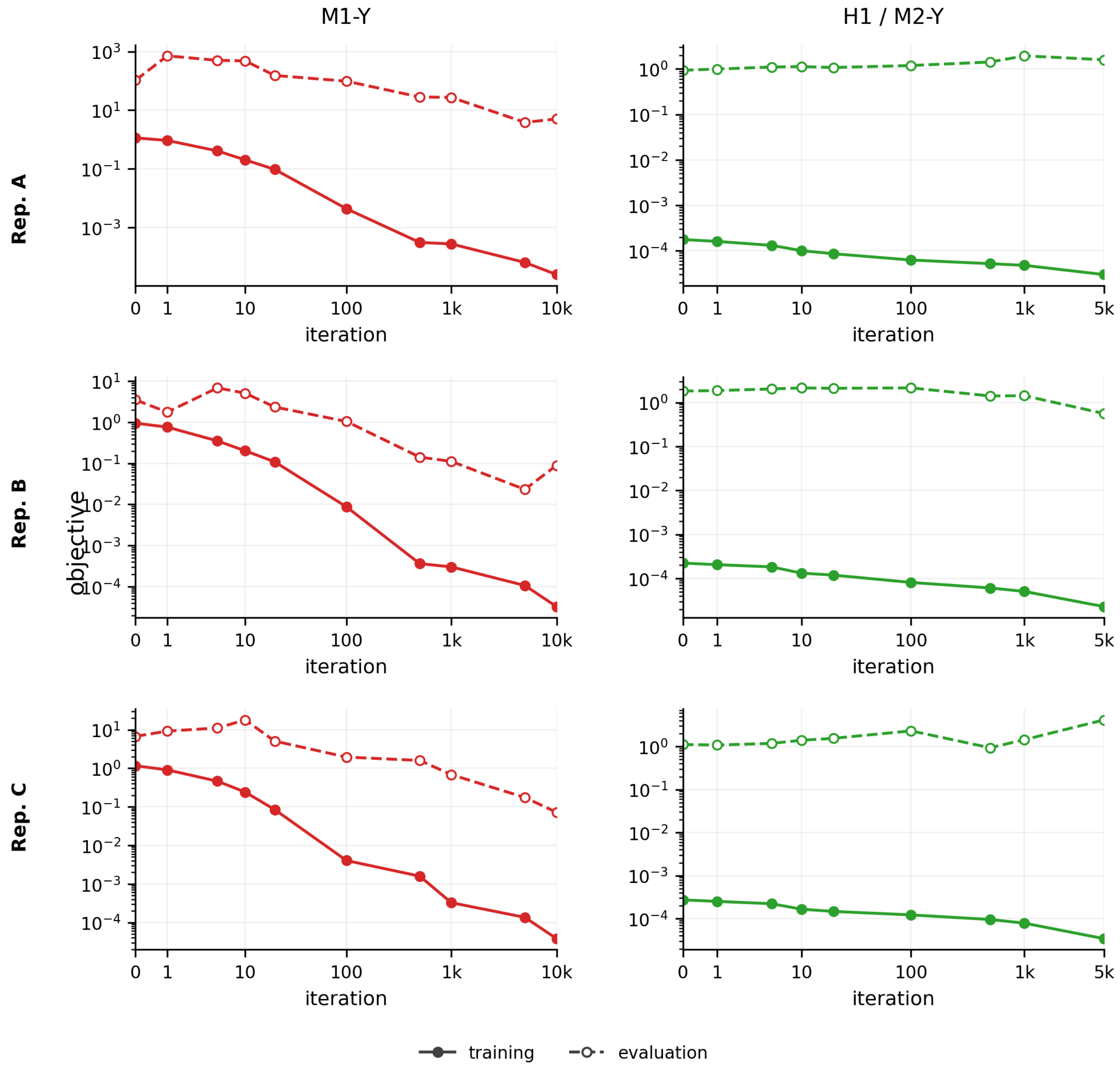

Figure 6: Training and evaluation errors for M1-Y and H1/M2-Y in Representations A–C. The left column shows the M1-Y direct physical-law objective evaluated at the pre-moist states $Y^{\star}$, and the right column shows the H1/M2-Y one-step state objective. Solid curves show the objective over the training portion of the trajectory, while dashed curves with open symbols show the same objective over the later evaluation portion. The evaluation data do not influence the optimization. Objective normalizations differ among representations and between M1-Y and H1/M2-Y, so numerical values should be interpreted within each panel rather than compared directly across panels.

The training–evaluation gap is substantial for both objectives and for all three representations. Thus a small training error alone is not sufficient to establish that the learned parameterization remains accurate as the system moves into later portions of the trajectory. We next compare the final networks using a common physical diagnostic: their ability to reproduce the local moist physics at the pre-moist state $Y^\star$.

4.4 Accuracy of the learned moist physics

The preceding comparison measures each training method using the objective for which it was optimized. We now compare the final networks using a common physical measure: how accurately they reproduce the moist-physics forcing, or the part of the forcing replaced by the network, when evaluated at the truth-derived pre-moist state $Y^\star$. Figure 7 shows this comparison for the final M1-Y, H1, H2, and H5 networks, separately over the training (filled symbols) and evaluation (open symbols) portions of the trajectory.

For Representation A, all four models reproduce the phase-change rate A accurately over the training interval, with relative RMS errors below 10^{-2}. The errors are substantially larger over the evaluation interval, where none of the models retains the same level of accuracy. H1, H2, and H5 nevertheless give smaller evaluation errors in A than M1-Y, with relative RMS errors of about 1.3 compared with approximately 2.24 for M1-Y. Thus, although H1 is not trained by directly minimizing the local error in A, the resulting H1, H2, and H5 networks give smaller later-time errors in this quantity than M1-Y. This comparison also illustrates why the objective values in Fig. 6 should not be used to rank M1-Y and H1 directly: they measure different quantities, whereas the present figure evaluates both networks using the same physical error in A.

A similar reduction in the evaluation error of A is found for Representation B. **The behavior of the learned rain rate R, however, is very different.** M1-Y reproduces the analytical rain law with relative RMS errors of only a few percent on both the training and evaluation intervals. In contrast, the H1, H2, and H5 models have errors of approximately 0.5 over the rain-active samples. Thus the models obtained from the state-based objectives give smaller later-time errors in A but substantially poorer recovery of the thresholded rain process.

Representation C shows an even stronger distinction. M1-Y closely reproduces the four-component moist source over the training interval and retains a source-vector relative RMS error of approximately 0.27 over the evaluation interval. The H1, H2, and H5 models have considerably larger source-vector errors, approximately 0.9 on the evaluation interval. Thus, although these models were trained to improve the evolution of the numerical state, they recover the underlying four-component moist source considerably less accurately than the direct physical-law regression.

These results show that reducing a state-based training objective does not necessarily improve the fidelity of every local physical process represented by the network. This distinction is particularly important for Representations B and C, where the network is given additional freedom to represent the rain process or the complete moist-source vector. We next examine whether these differences in local physical accuracy lead to corresponding differences when the learned parameterizations are actually deployed in the DIMSWE simulation.

4.5 Physical behavior after deployment

We next examine the learned parameterizations after they are embedded in the DIMSWE timestep and allowed to evolve the model state. Figure 8 summarizes several diagnostics of the final deployed solutions for M1-Y, H1, H2, and H5.

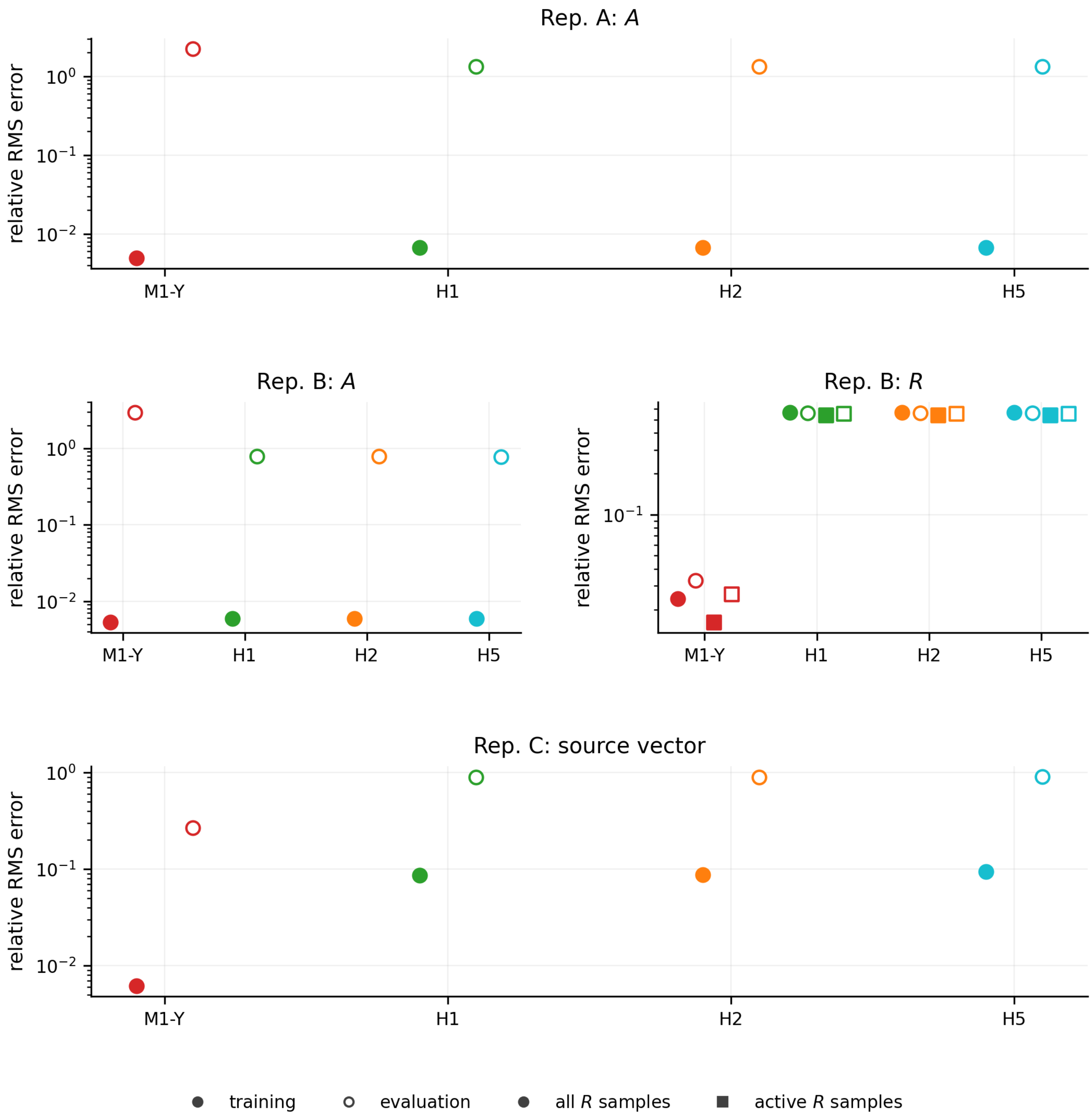

Figure 7: Accuracy of the final learned moist-physics models evaluated at the pre-moist truth states $Y^\star$. Filled symbols correspond to training states 0–80 and open symbols to evaluation states 81–160. For Representation A, the error is measured in the phase-change rate A. For Representation B, separate panels show errors in A and R; the R panel distinguishes errors over all samples and over samples for which the analytical rain rate is active. For Representation C, the plotted quantity is the normalized source-vector relative RMS error. $Y^\star$ is the truth-derived pre-moist state corresponding to the moist-physics call site in the split timestep.

The three representations behave quite differently with respect to conservation. Representations A and B preserve the total-water and thermodynamic source identities by construction, and their total-water drift remains at approximately roundoff level for all four training objectives. Representation C does not impose these identities, and its total-water drift increases along the sequence M1-Y $\rightarrow$ H1 $\rightarrow$ H2 $\rightarrow$ H5.

The final state errors show a similar pattern. Representations A and B remain accurate for all four training objectives, with relative errors of order 10^{-5} or smaller. Representation C is also accurate for M1-Y, but its final state error increases markedly for H1, H2, and H5, reaching order 10^{-3} for H5. **Thus, for the unconstrained source representation, increasing the complexity of the training objective (which requires online training) does not improve long-time deployment.**

None of the representations explicitly enforces non-negativity of the discrete moisture fields. Negative values occur in the discrete moisture variables Q_v, Q_c, and Q_r for all three representations, with substantially larger violations for Representation C after H1, H2, and H5 training. The rain and cloud mass diagnostics show a similar distinction. Representations A and B retain comparatively small errors in the final water partition, whereas Representation C develops substantially larger cloud- and rain-mass errors, particularly for the state-based and recursive objectives.

Rain onset provides an additional diagnostic of the learned rain process. Representation A retains the analytical rain law and reproduces the truth onset time. In Representations B and C, the learned rain tendency is not constrained to reproduce the analytical threshold exactly, and nonzero rain production appears before the truth onset. Because an arbitrarily small positive learned tendency can trigger this timing diagnostic, the onset error is best interpreted together with the rain-rate and water-partition errors rather than as a measure of rain accuracy by itself.

4.6 Global evolution of the deployed models

The endpoint diagnostics in Fig. 8 summarize the final accuracy of the deployed models, but do not show how the differences develop during the simulation. Figures 9–11 therefore compare several domain-integrated and flow diagnostics throughout the full DoubleVortex trajectory.

For Representation A, all four learned models reproduce the cloud- and rain-water evolution closely (Fig. 9). The total-water drift remains at roundoff level, as expected from the prescribed source structure. Differences among the training objectives are more apparent in the flow diagnostics: M1-Y remains relatively close to the truth in kinetic energy and projected relative-vorticity squared, while H1, H2, and H5 develop somewhat larger deviations late in the simulation. Nevertheless, the overall state errors remain small for all four models. Thus, when only the phase-change rate A is learned and the analytical rain law and source structure are retained, the deployed solution is comparatively insensitive to the choice among the four training objectives.

Representation B remains numerically stable and also preserves total water to roundoff, but the rain-water evolution is more sensitive to the training objective (Fig. 10). M1-Y slightly overpredicts the final rain-water reservoir but remains closest to the truth, whereas H1, H2, and H5 underpredict it. This behavior is consistent with the local-physics results above: M1-Y provides the most accurate approximation of the analytical rain rate R, while the state-based models tolerate substantially larger local errors in R while still maintaining small overall state errors. The kinetic-energy and vorticity diagnostics remain close to truth for all four models. Thus an accurate global state trajectory does not require equally accurate recovery of each individual local moist-process rate.

Representation C behaves qualitatively differently (Fig. 11). M1-Y remains comparatively close to the truth, although its water partition and total-water conservation are less accurate than in the structured representations. The H1, H2, and H5 models progressively depart from the physical solution. Cloud water is increasingly underestimated, while the predicted rain-water reservoir

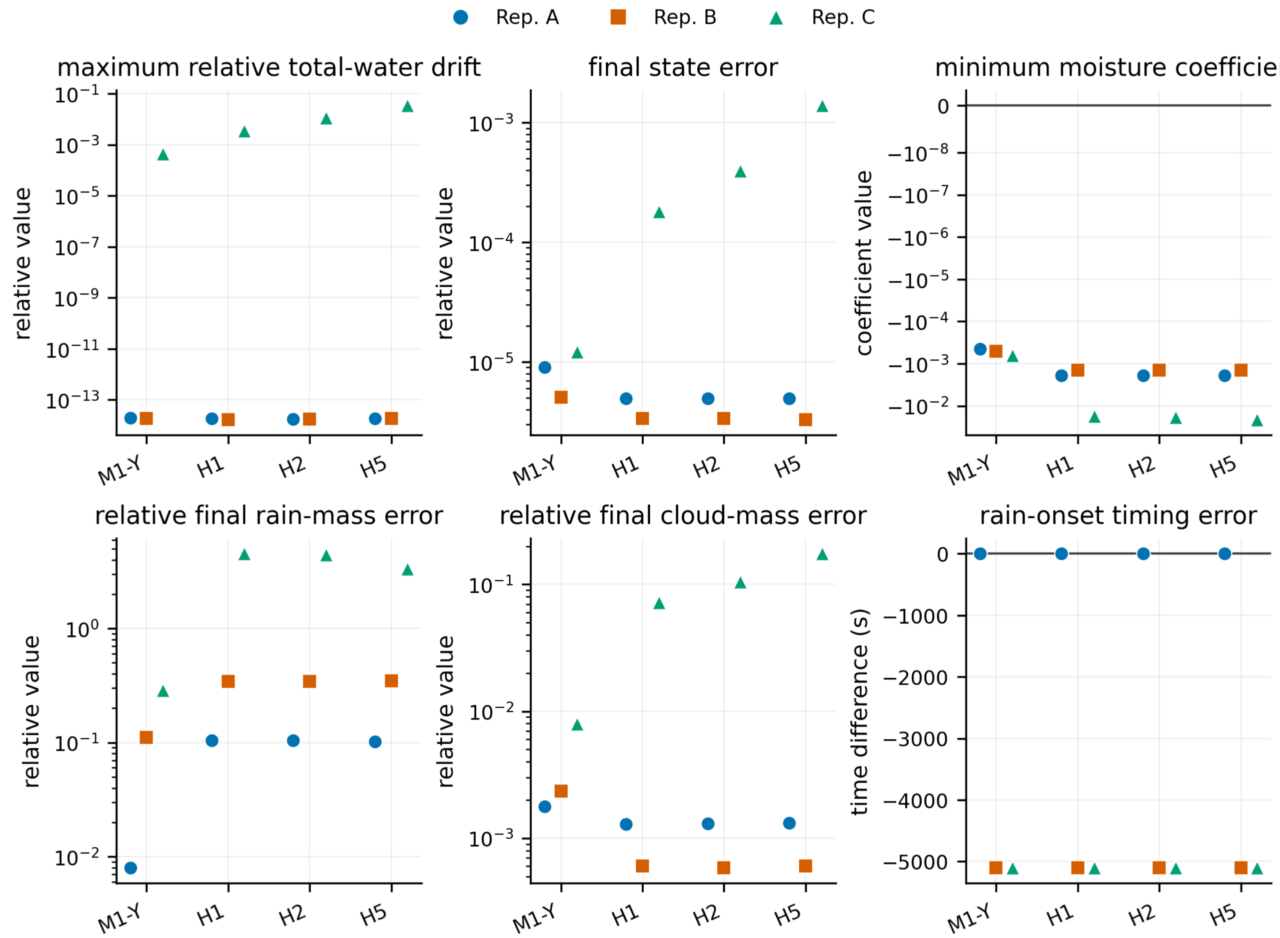

Figure 8: Physical diagnostics of the deployed Test 2B models for Representations A–C. The panels show the maximum relative total-water drift, final relative state error, minimum moisture coefficient, relative errors in the final rain- and cloud-water masses, and the difference in rain-onset time relative to truth. Representations A and B enforce the total-water and thermodynamic source identities algebraically, whereas Representation C learns the four source components independently; the corresponding conservation errors therefore have different interpretations. The reported moisture minima are finite-element coefficient minima and should not be interpreted as a pointwise positivity guarantee for the reconstructed continuous fields.

becomes negative and its magnitude grows during the simulation. These models also develop substantial total-water drift and considerably larger errors in the kinetic-energy and relative-vorticity-squared diagnostics. The final state error increases from M1-Y through H1, H2, and H5.

This behavior is notable because the H2 and H5 training errors decrease during their respective optimization stages (Fig. 5), while the error after the full 160-step deployed simulation increases. Thus, for Representation C, improving the model over the two- and five-step training windows does not lead to a more accurate long-time solution. Because H2 and H5 were each optimized for only 20 iterations, however, the present calculations do not determine whether additional optimization would alter this behavior.

The Representation-C results demonstrate the importance of distinguishing trajectory optimization from physical structure. Because the four moist-source components are predicted independently, neither water conservation nor the physical cloud–rain partition is guaranteed. The H1, H2, and H5 sequence does not recover these relationships and, for the present case, produces

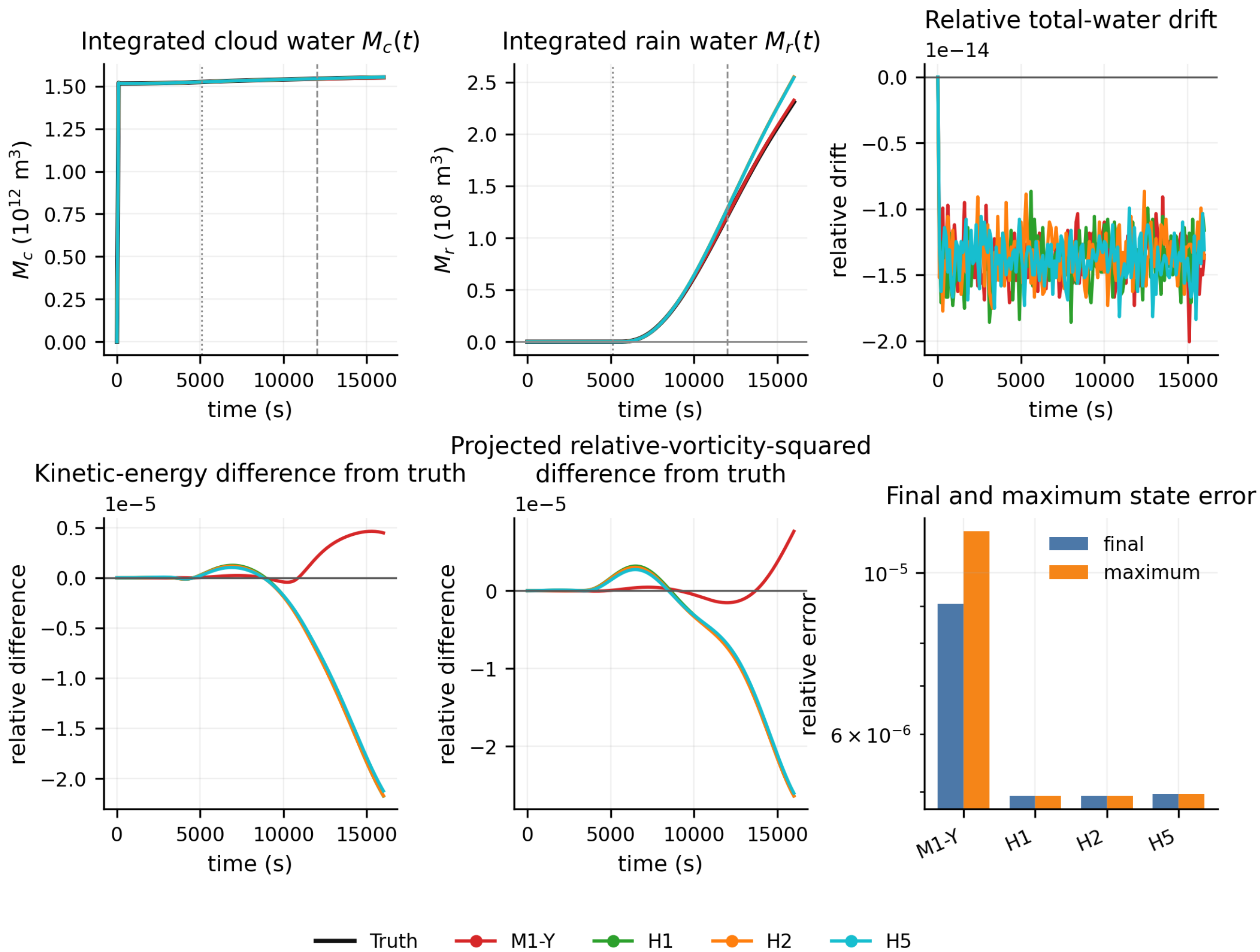

Figure 9: Global evolution of the deployed Representation-A models. The panels show integrated cloud water M_c, integrated rain water M_r, relative total-water drift, the difference in kinetic energy from truth, the difference in projected relative-vorticity squared from truth, and the final and maximum relative state errors. The vertical lines indicate first truth rain production and the time of peak integrated truth rain production. Total-water conservation is imposed by the Representation-A source structure.

a numerically well-defined but progressively less physical evolution.

5 Deployed Hybrid Dynamics

The preceding section quantified the deployed models using local errors, conservation diagnostics, and domain-integrated trajectories. We now illustrate the evolution of corresponding moisture fields. All truth and learned-model plots use the same five reference times and the same color limits for each variable.

5.1 Truth

Figure 12 provides the common truth reference. The columns correspond to t = 0, 5000, 6100, 12000, and 16000 s. The rows show relative humidity, specific cloud water q_c, specific rain water q_r, and rain-production rate R. Gray contours indicate relative vorticity.

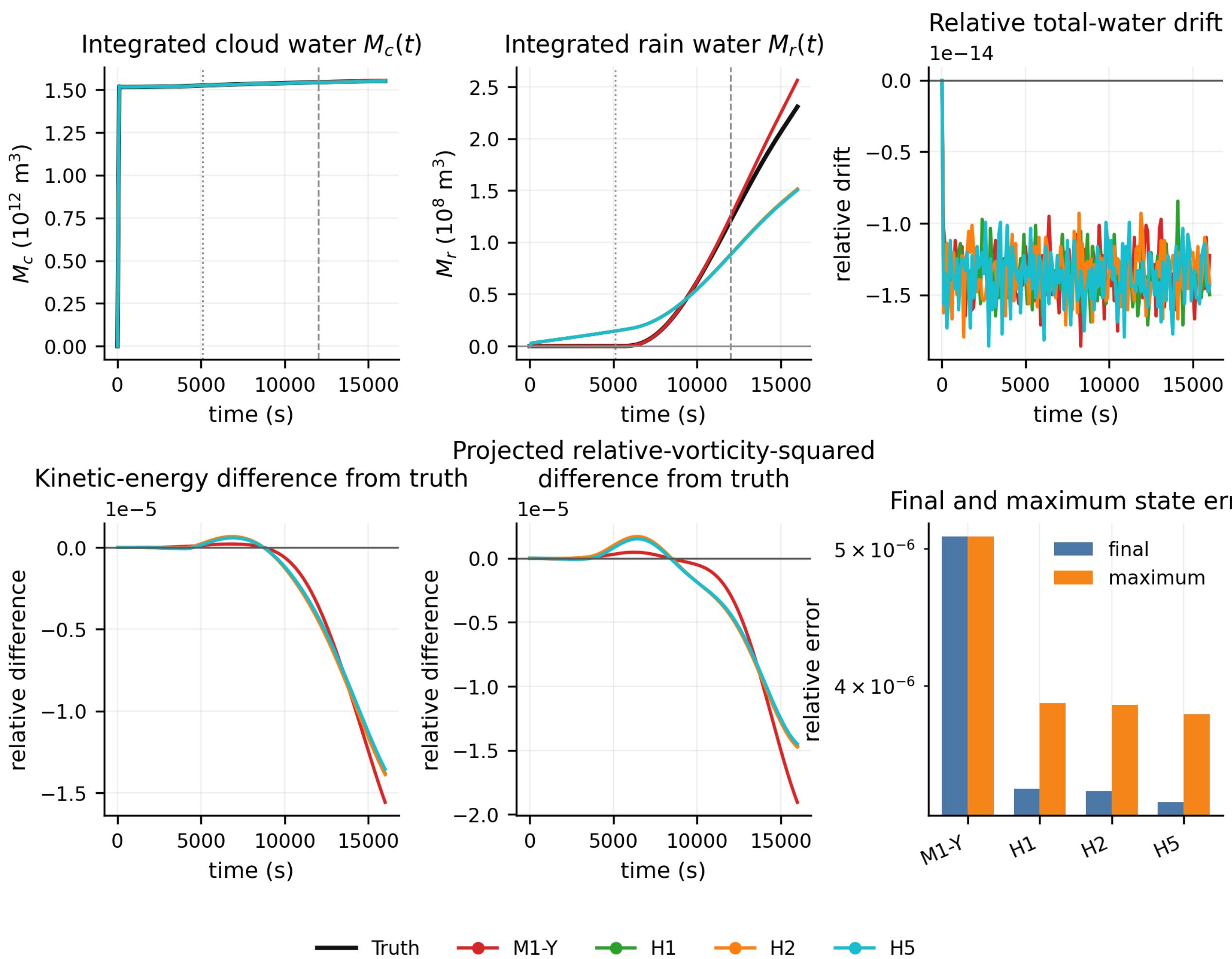

Figure 10: Global evolution of the deployed Representation-B models. The panels show integrated cloud water M_c, integrated rain water M_r, relative total-water drift, the difference in kinetic energy from truth, the difference in projected relative-vorticity squared from truth, and the final and maximum relative state errors. Representation B learns both A and R while retaining the prescribed moist-source structure, so total-water conservation remains enforced algebraically.

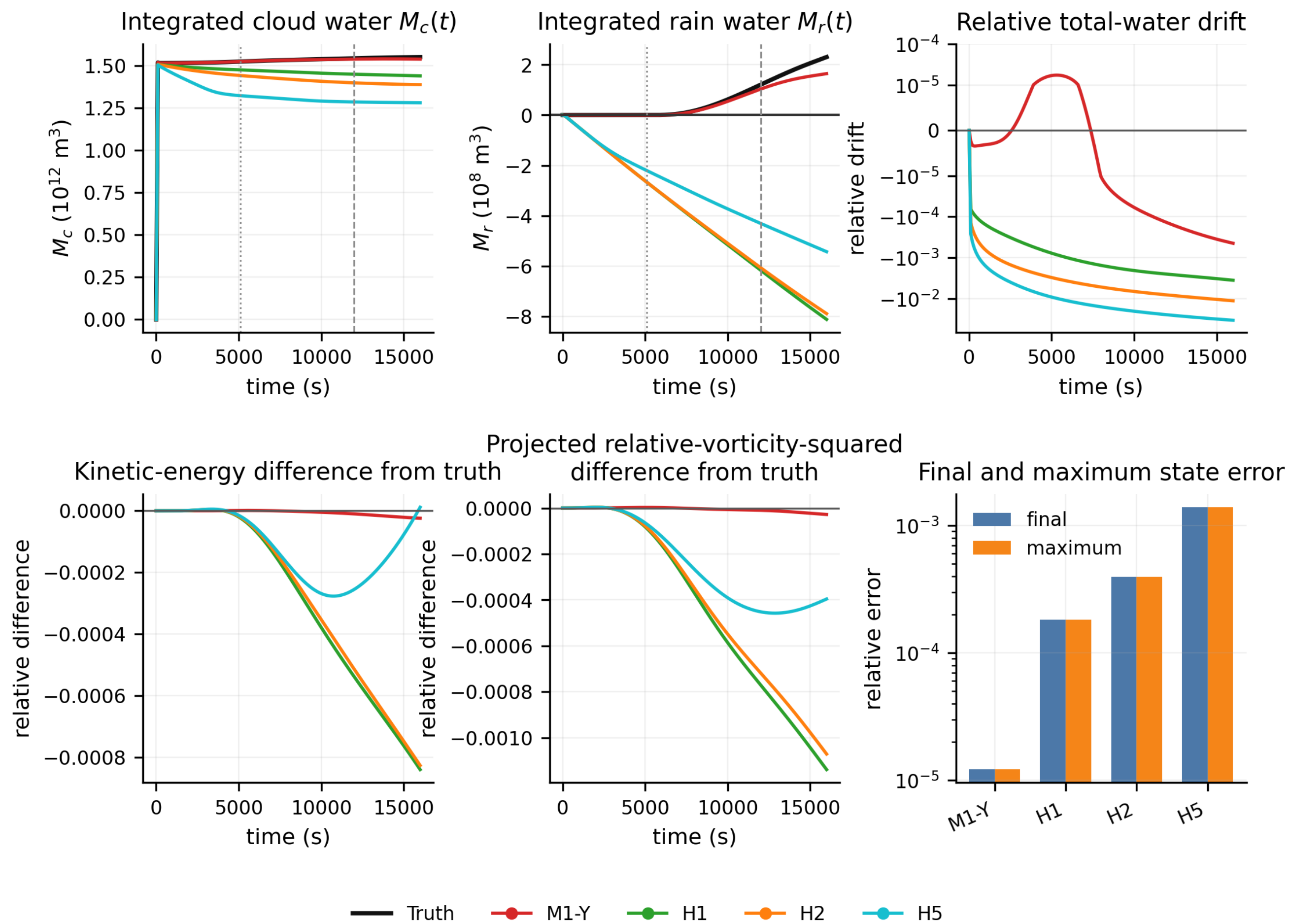

Figure 11: Global evolution of the deployed Representation-C models. The panels show integrated cloud water M_c, integrated rain water M_r, relative total-water drift, the difference in kinetic energy from truth, the difference in projected relative-vorticity squared from truth, and the final and maximum relative state errors. Unlike Representations A and B, Representation C predicts the four moist-source components independently and does not impose total-water conservation or the analytical cloud–rain source structure. The horizontal zero line in the rain-water panel highlights the negative rain water produced by several of the state-based models.

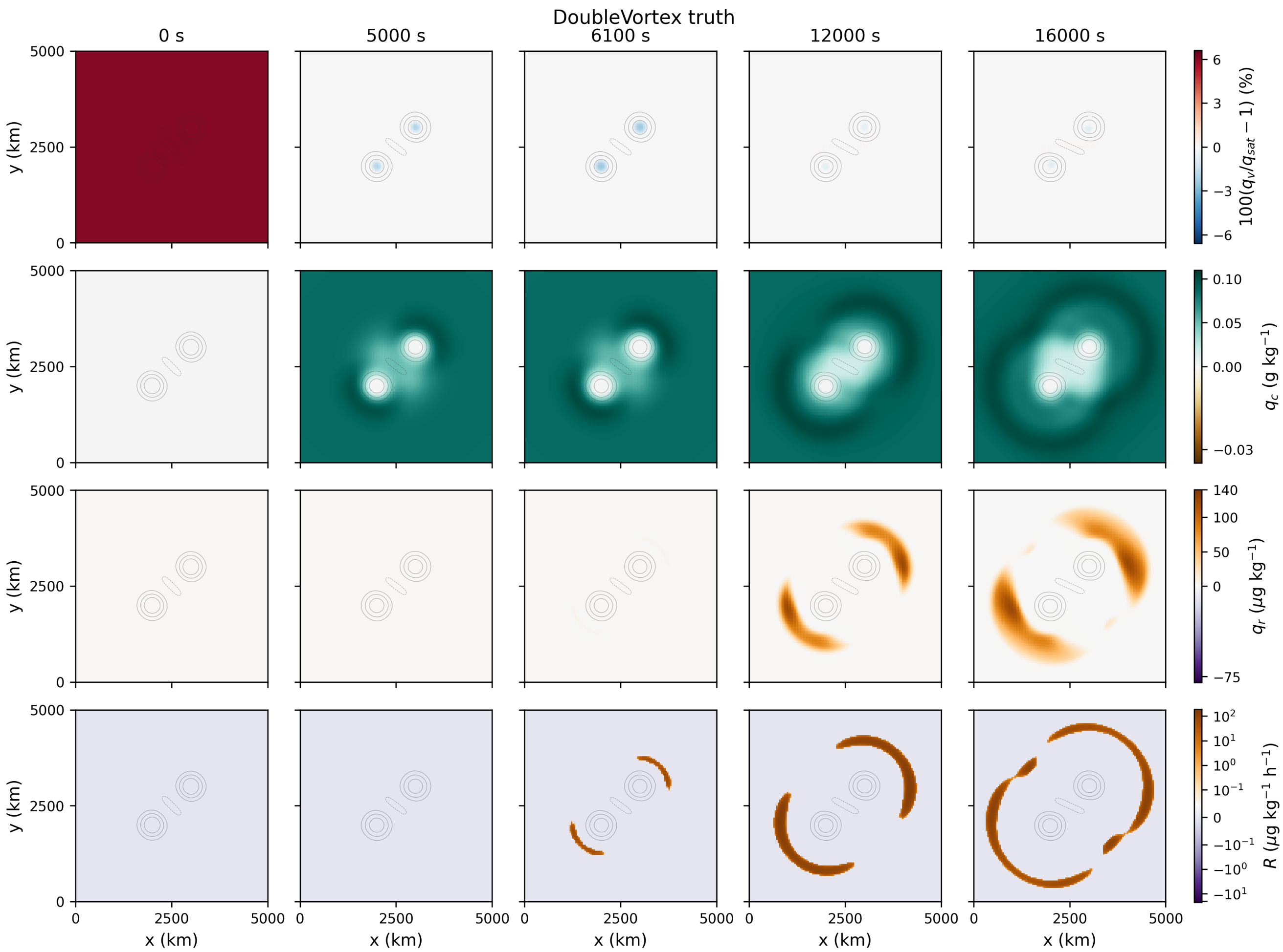

Figure 12: Common spatial reference for the ground-truth DoubleVortex evolution. Columns show t = 0, 5000, 6100, 12000, and 16000 s. Rows show saturation departure $100(q_v/q_{\text{sat}} - 1)$, specific cloud water q_c, specific rain water q_r, and the analytical rain-production rate R, respectively. Gray contours show relative vorticity. The same color limits, spatial coordinates, and vorticity contour levels are used in all deployed-model galleries.

5.2 Representation A

The deployed evolution is robust across all four training objectives illustrated across figures 13 to 16. The two-vortex structure is preserved and the location and overall shape of the cloud and rain bands is close to that in the truth solution. Minute differences arise in amplitude and in the detailed late-time relative humidity and rain-production fields. Overall, these spatial results are consistent with the small state and water-partition errors found in Sec. 4.

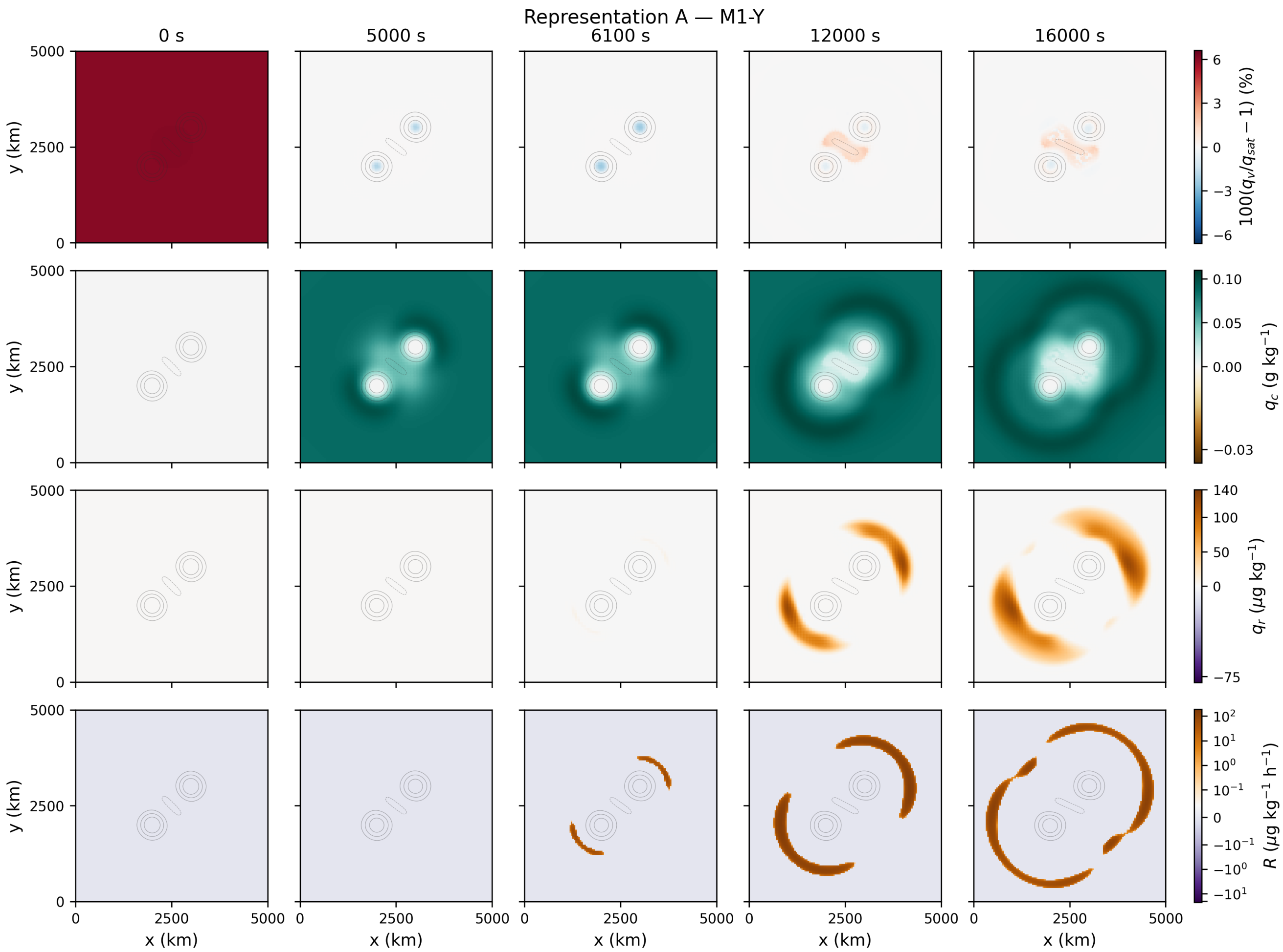

Figure 13: Same as figure 12 but for Representation-A evolution using M1-Y.

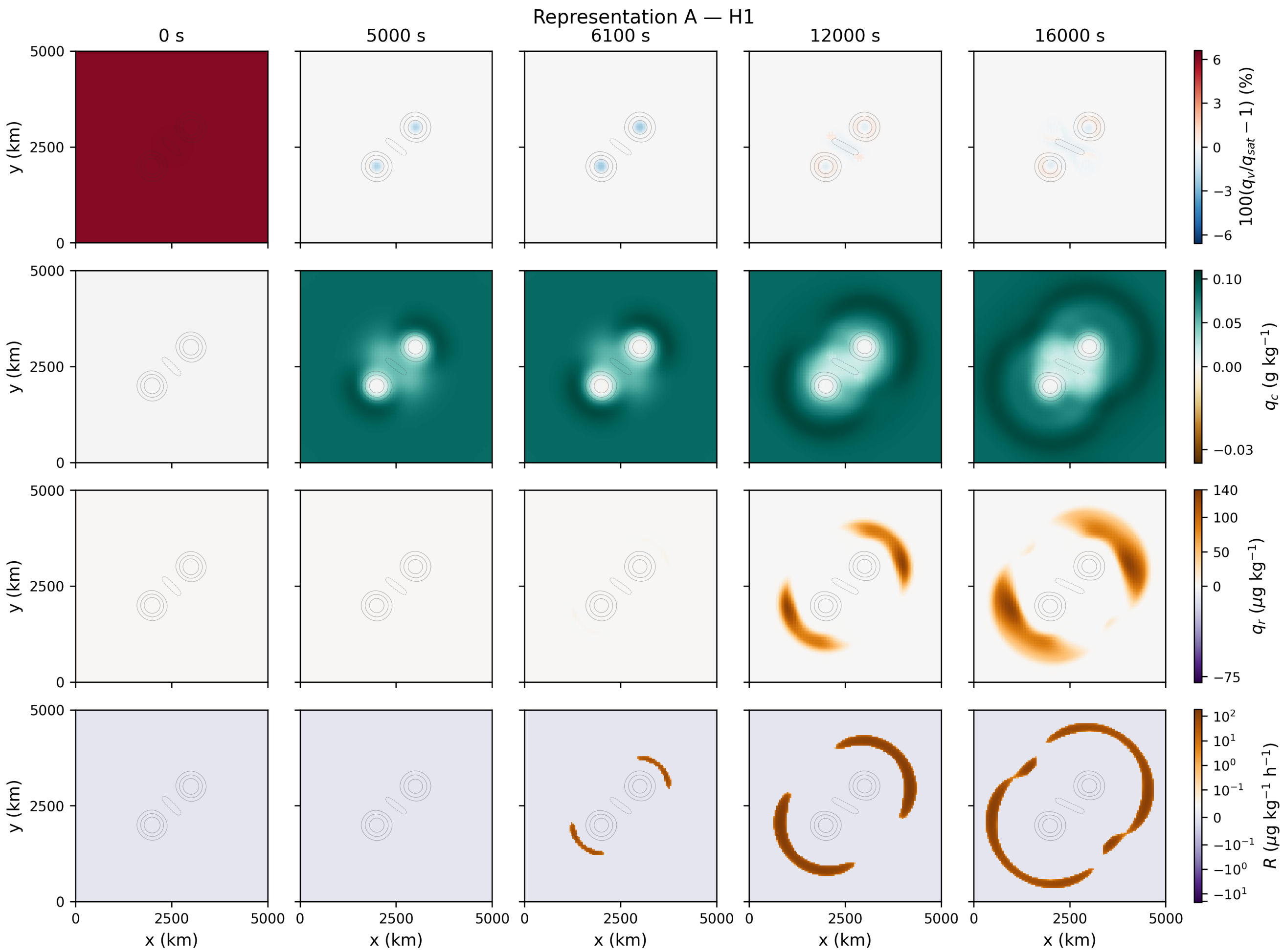

Figure 14: Same as figure 12 but for Representation-A evolution using H1/M2-Y.

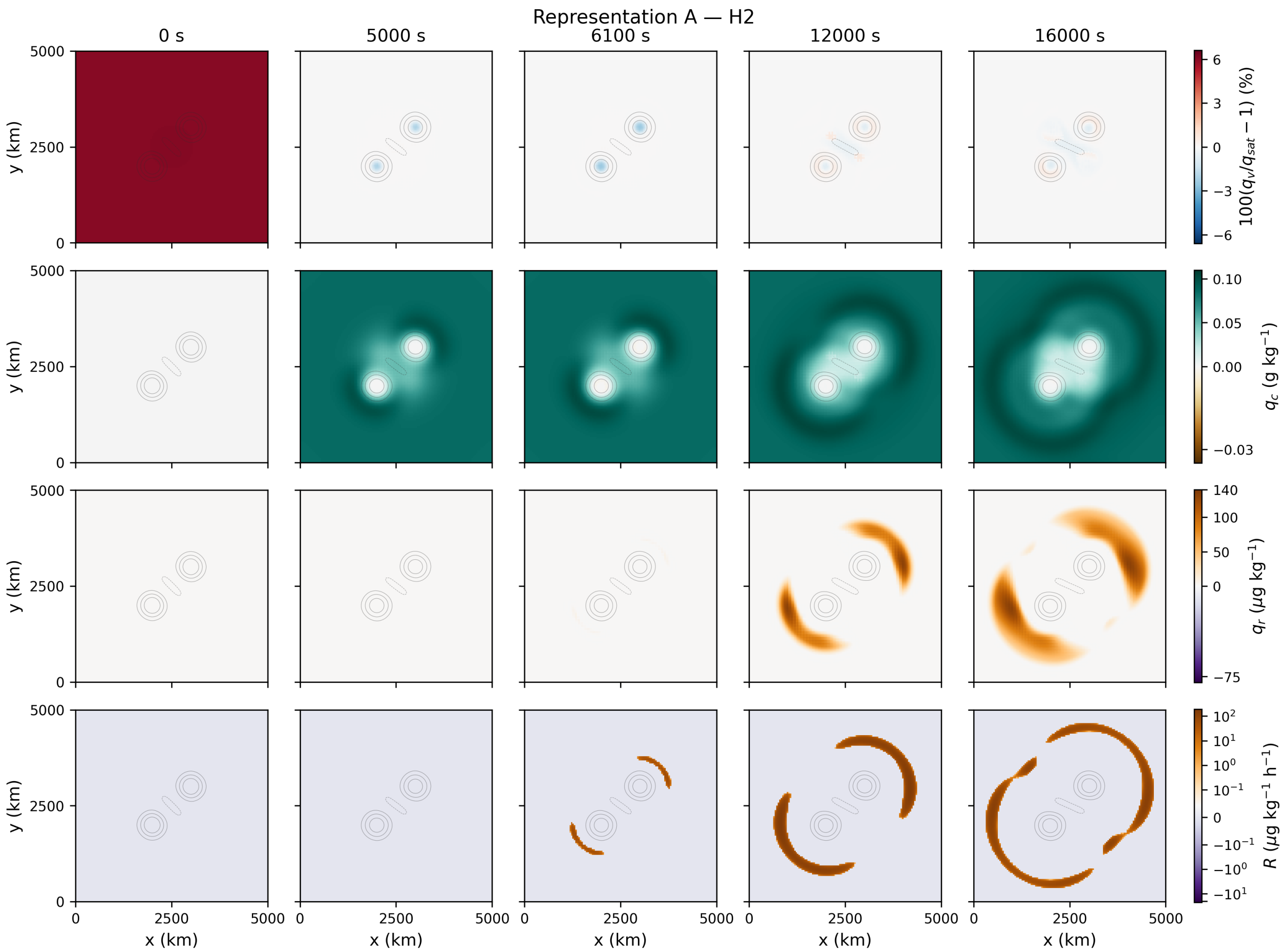

Figure 15: Same as figure 12 but for Representation-A evolution using H2.

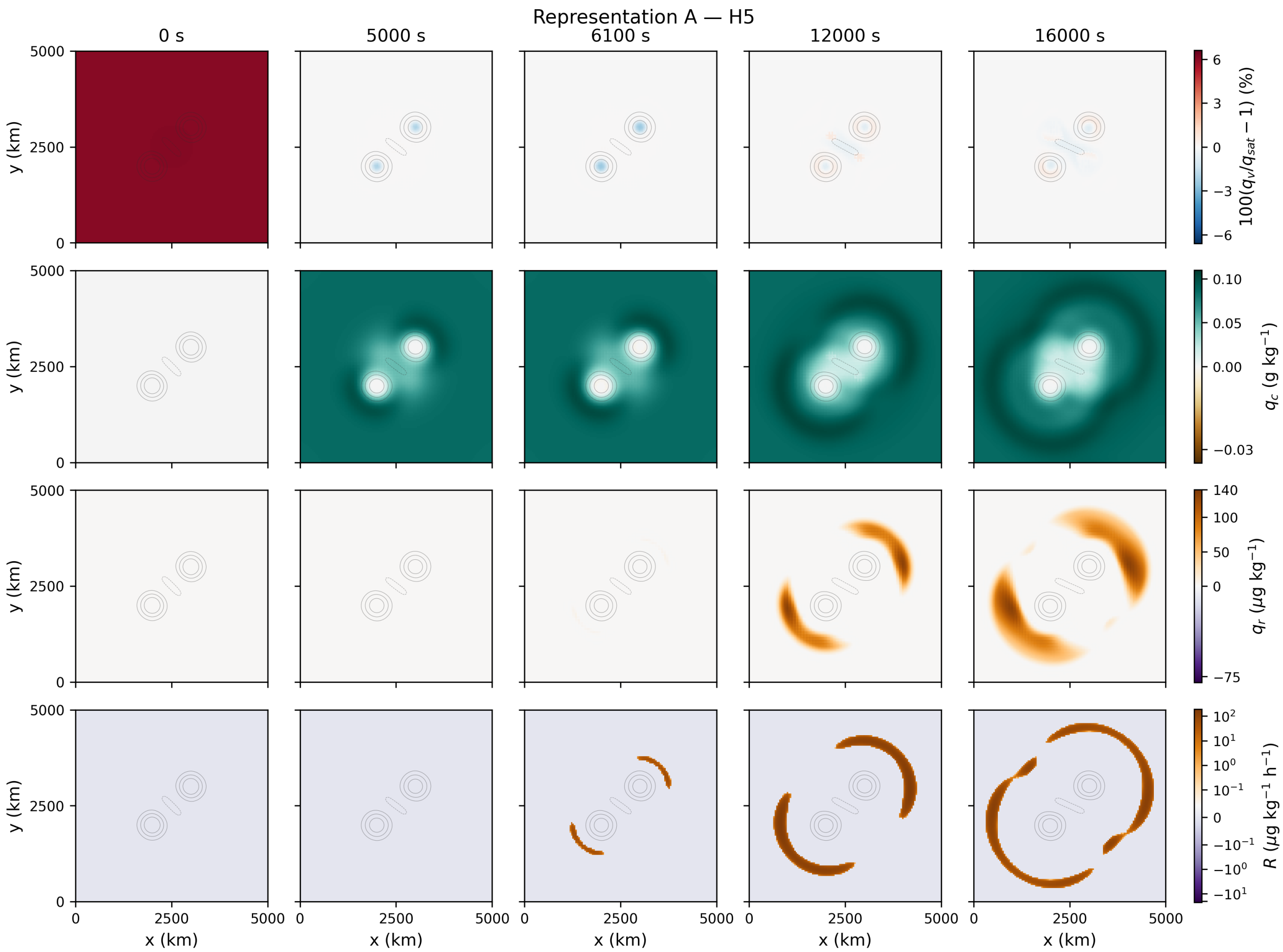

Figure 16: Same as figure 12 but for Representation-A evolution using H5.

5.3 Representation B

Representation B also preserves the principal spatial structure of the solution. The vortex skeleton, cloud envelope, and locations of the two rain bands remain close to truth for all four training objectives. M1-Y produces rain-water bands closest to the truth (figure 17), whereas H1 (figure 18), H2(figure 19), and H5(figure 20) retain the same basic morphology but noticeably underpredict their strength. The latter three are similar to each other.

The learned rain-production fields reveal much larger differences than the state variables themselves. M1-Y retains relatively localized positive and negative structures around the evolving rain bands. In contrast, H1, H2, and H5 develop broad regions of negative learned R surrounded by narrower positive structures, with nonzero activity already present before the truth rain onset. Thus the poorer local recovery of R identified in Fig. 7 is clearly visible in space even though the cloud fields, vortex structure, and overall deployed state errors remain comparatively small.

This provides a useful example in which the state evolution remains accurate while the learned representation of an individual physical process is substantially less faithful despite preserving interspecies conservation.

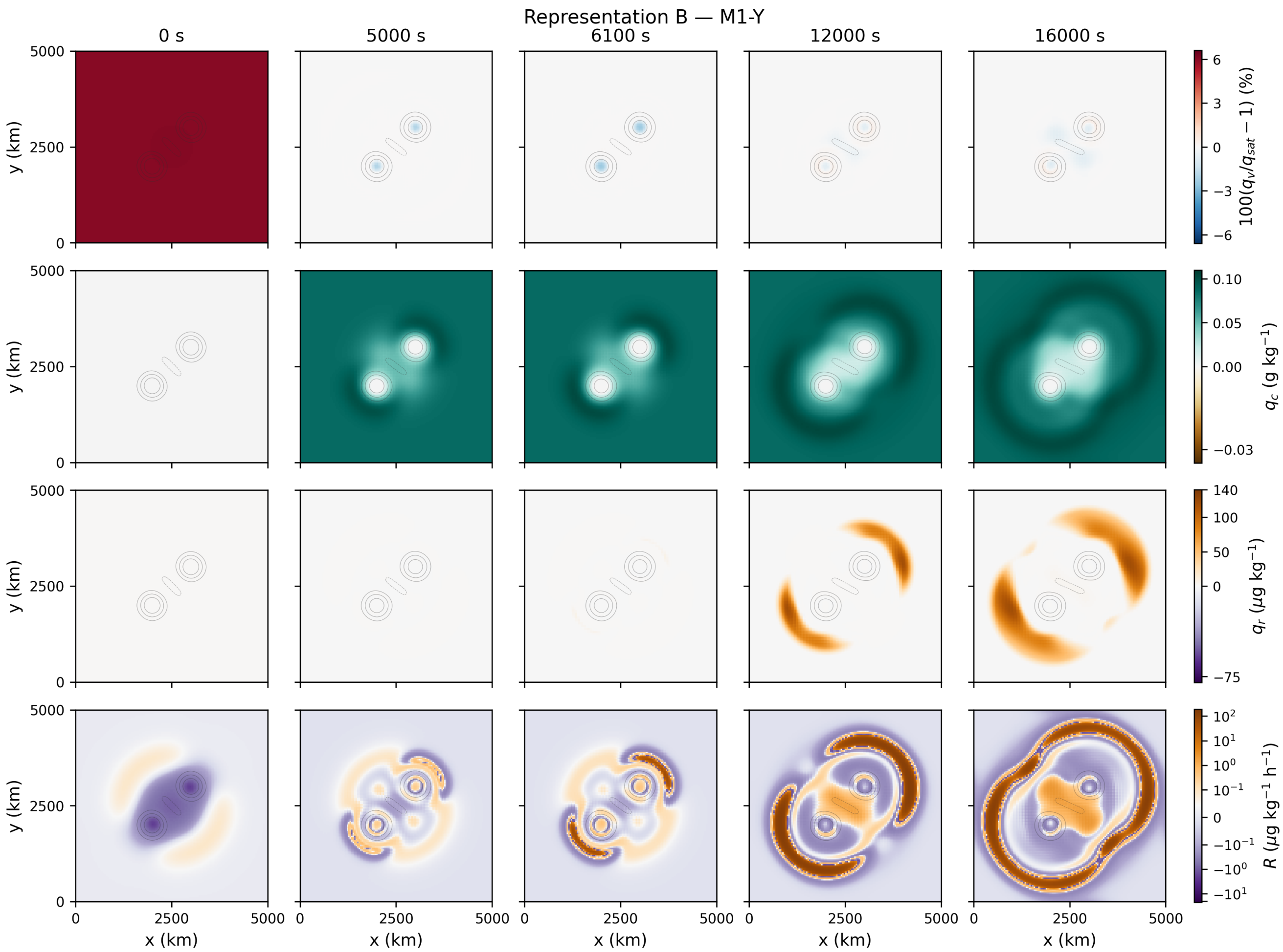

Figure 17: Same as figure 12 but for Representation-B evolution using M1-Y.

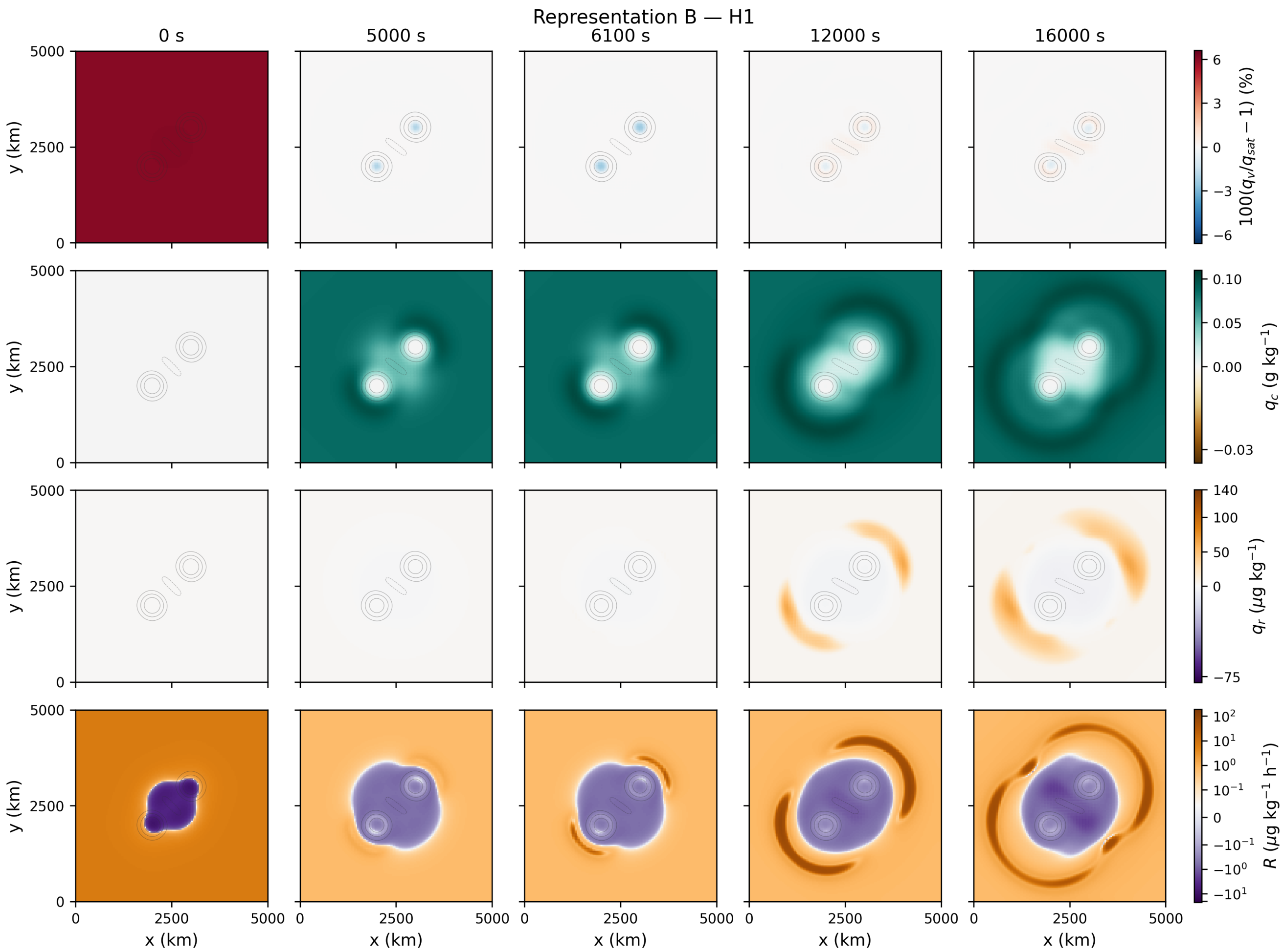

Figure 18: Same as figure 12 but for Representation-B evolution using M2-Y/H1.

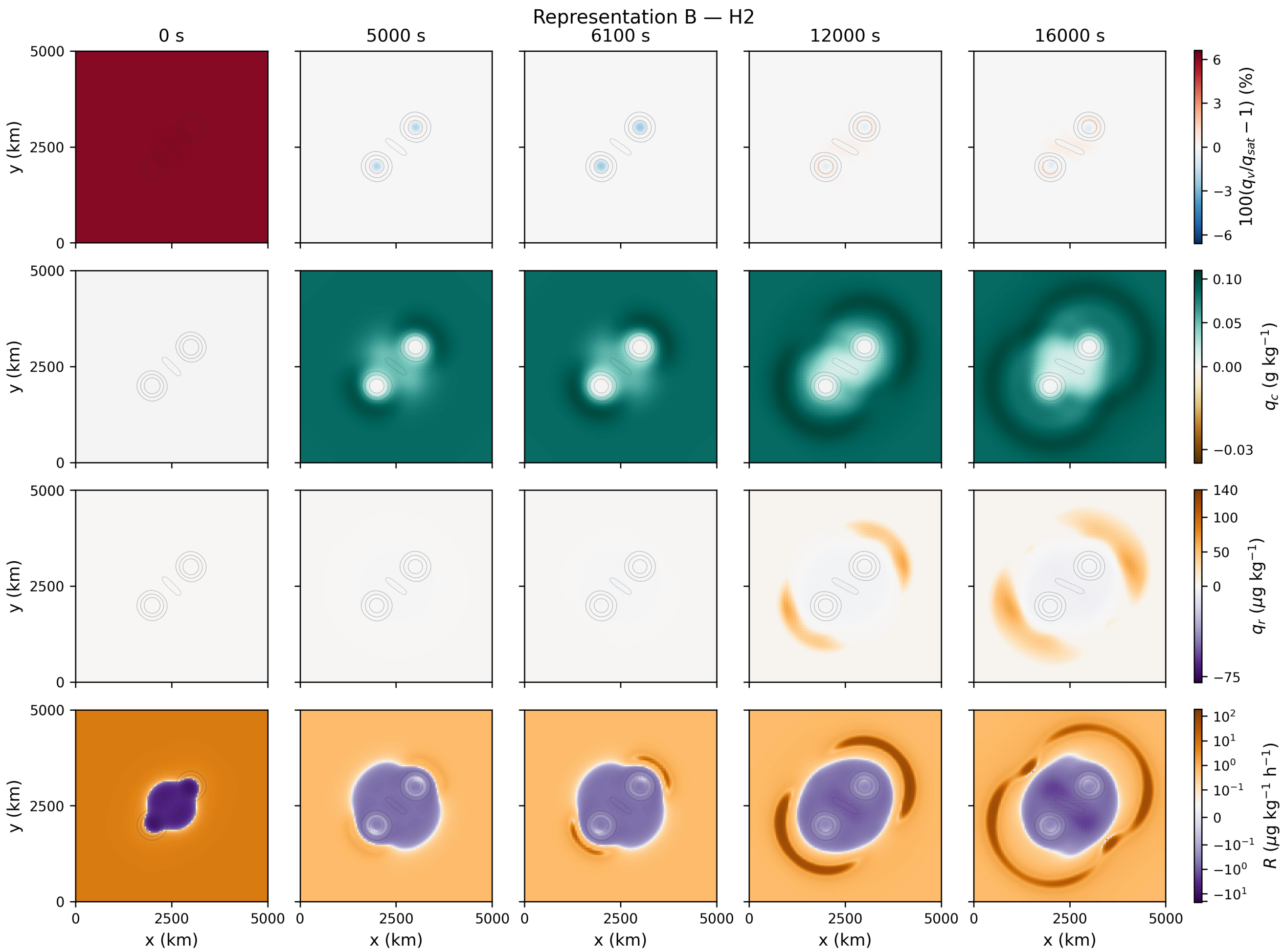

Figure 19: Same as figure 12 but for Representation-B evolution using H2.

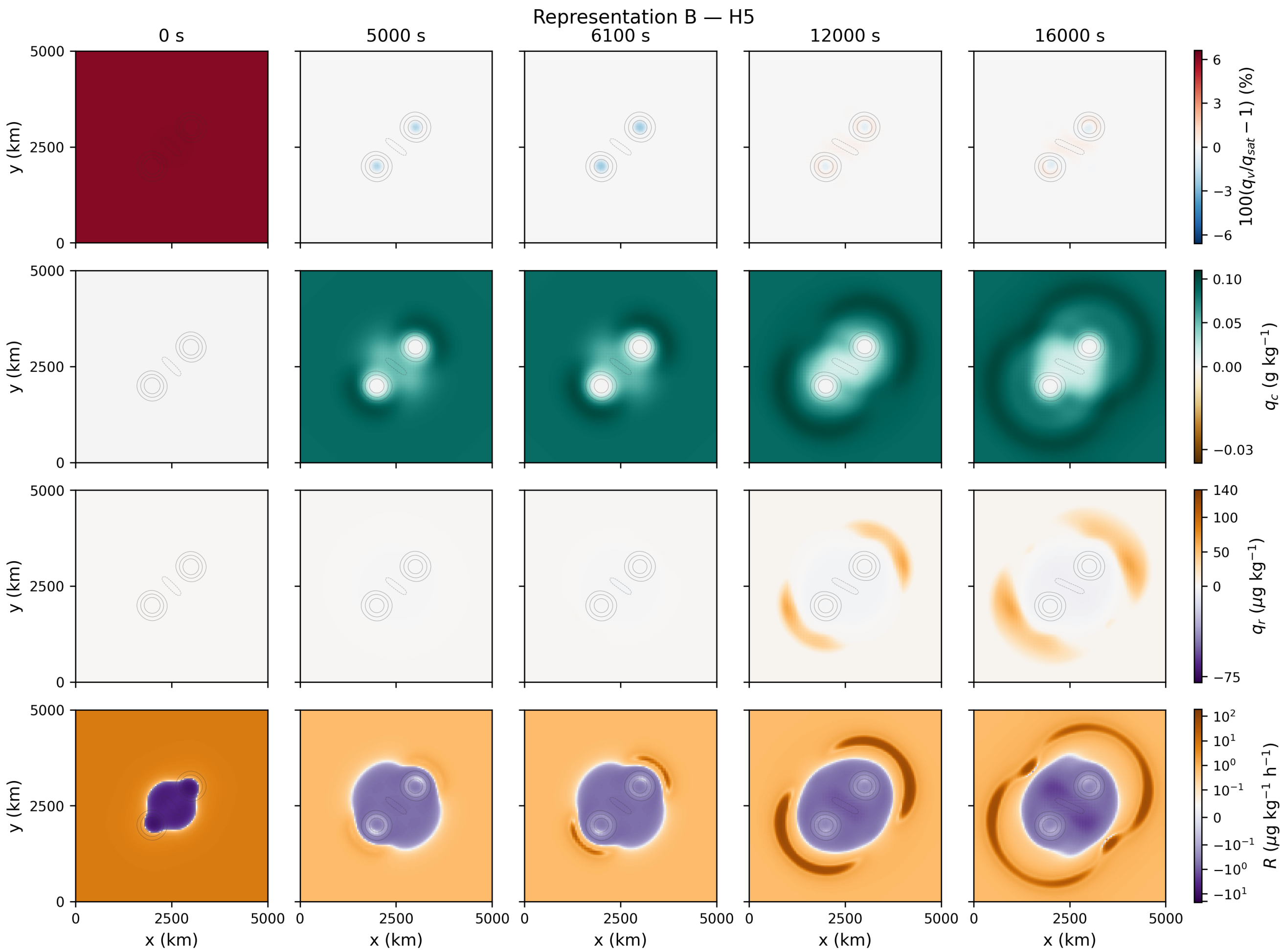

Figure 20: Same as figure 12 but for Representation-B evolution using H5.

5.4 Representation C

Representation C produces a qualitatively different outcome. The M1-Y model (figure 21) retains broadly truth-like cloud and positive rain-band morphology through the simulation, although its effective rain-production field is spatially oscillatory and contains both positive and negative structures. Thus a reasonably accurate state evolution can coexist with substantial freedom in the underlying unconstrained source representation.

The H1 (figure 22)), H2 (figure 23)), and H5 figure 24) models, however, develop a distinctly unphysical water partition. Negative rain water appears near the vortex cores by the 5000 s reference time and subsequently grows into broad core-centered negative structures rather than the positive outer rain bands seen in the truth solution. Negative cloud-water regions and stronger subsaturation accompany this behavior. The same basic failure appears in H1 and H2 and becomes most pronounced for H5. This is consistent with the conservation and water-partition diagnostics in Sec. 4: once the known two-rate source structure is removed, the H1, H2, and H5 training sequence does not recover the missing physical constraints.

Despite these large moisture errors, the relative-vorticity contours remain visually close to the common two-vortex skeleton. The deterioration is therefore concentrated primarily in the learned moist physics rather than in a wholesale loss of the underlying flow structure. At least, keeping dynamical core non-ML has a value in this case.

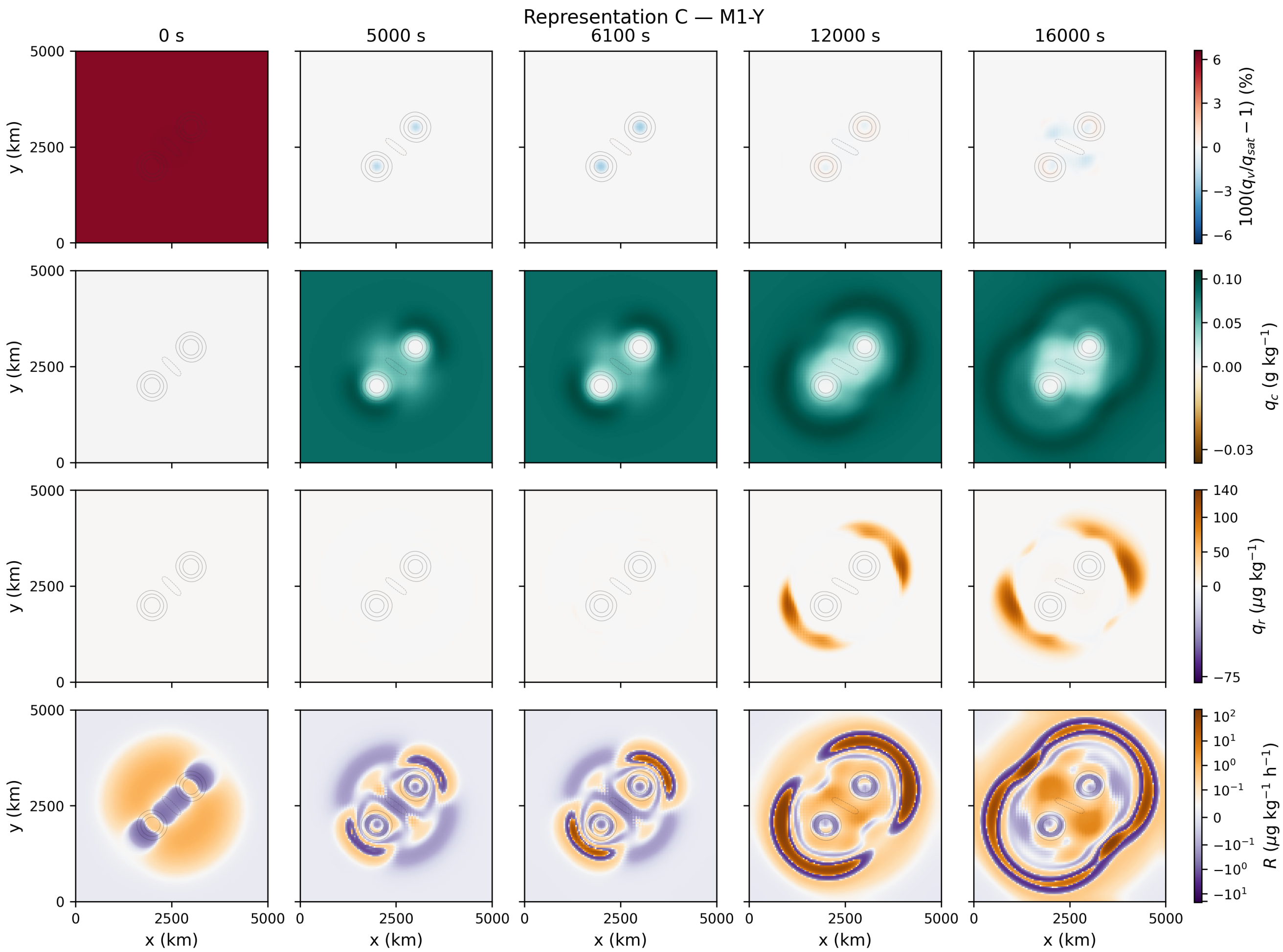

Figure 21: Same as figure 12 but for Representation-C evolution using M1-Y.

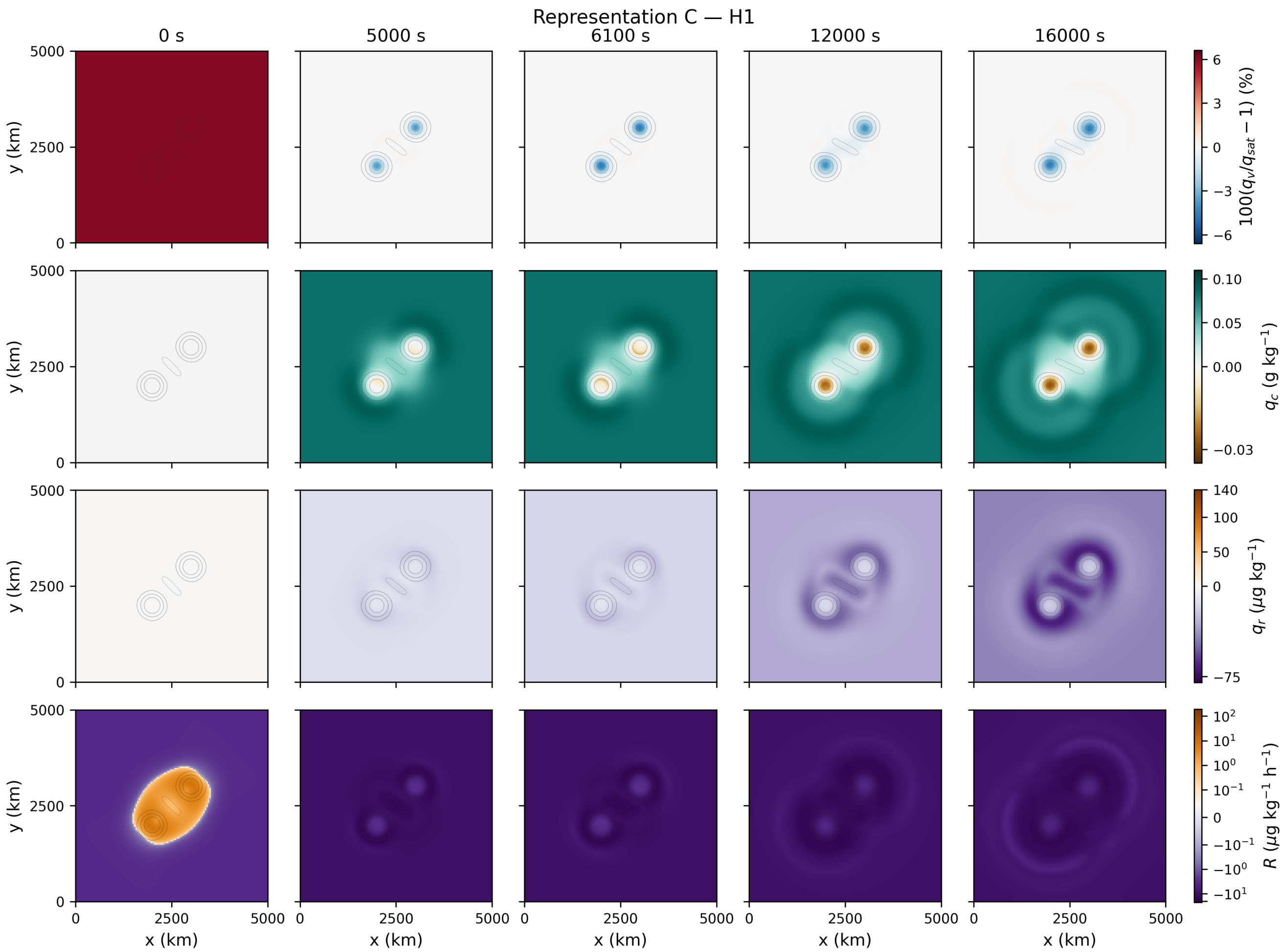

Figure 22: Same as figure 12 but for Representation-C evolution using M2-Y/H1.

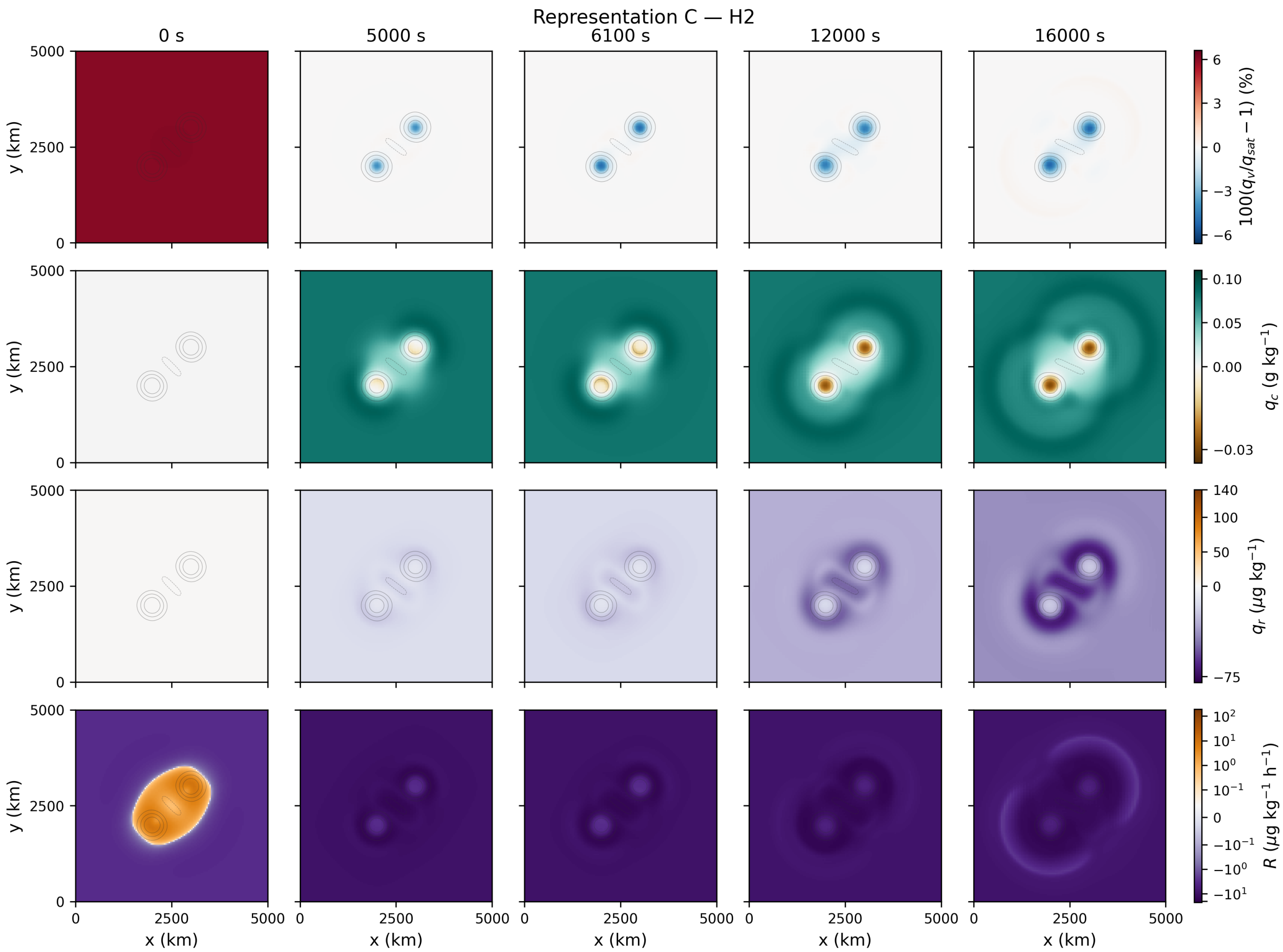

Figure 23: Same as figure 12 but for Representation-C evolution using H2.

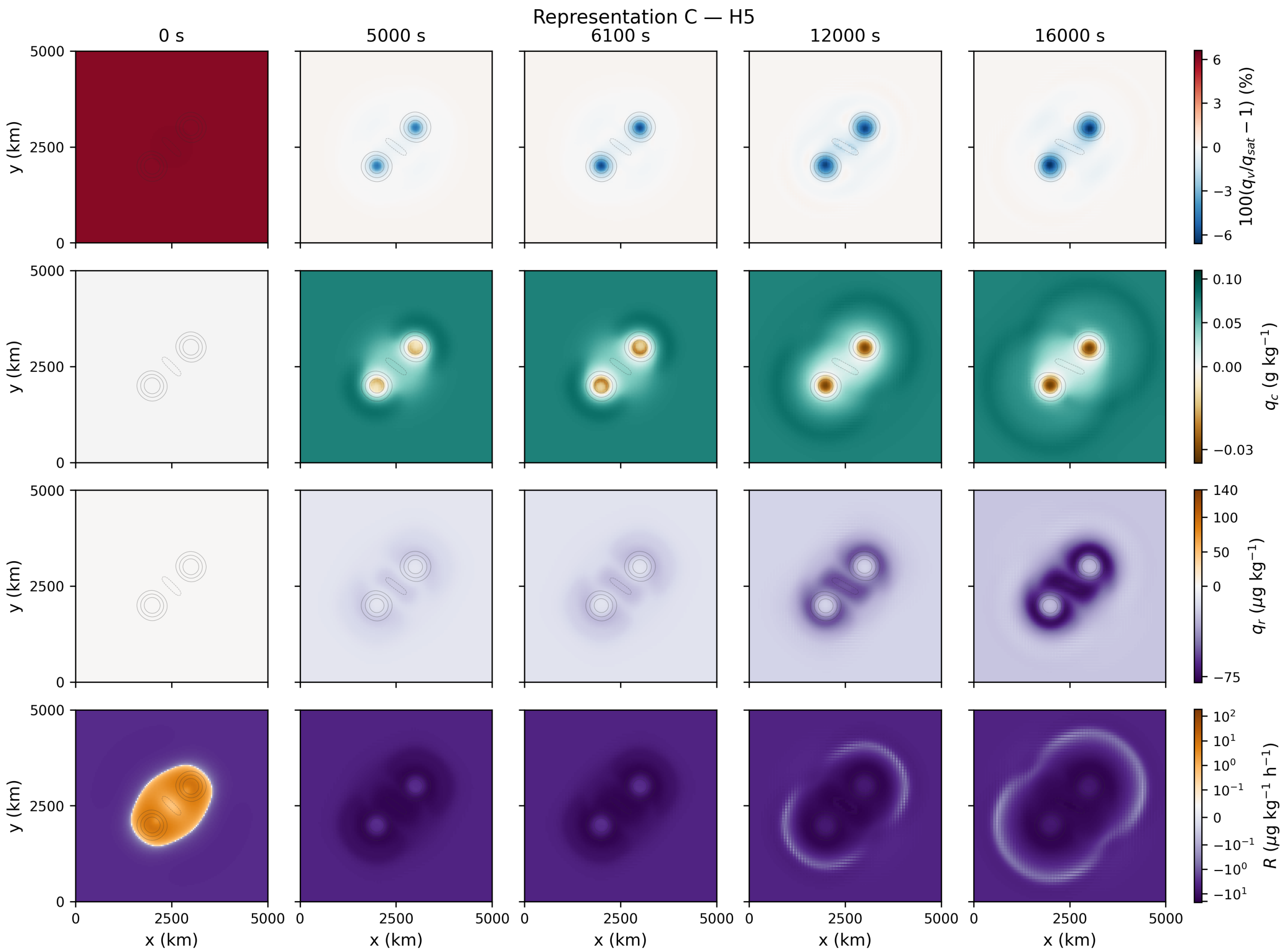

Figure 24: Same as figure 12 but for Representation-C evolution using H5.

The corresponding movies contain all 161 stored states and use the same fixed color limits as the galleries. They provide the continuous-time view of these dynamics and confirm the qualitative behavior identified above: Representation A remains spatially robust across the four training objectives, Representation B preserves the state morphology despite substantial differences in the learned rain-production field, and Representation C develops increasingly unphysical moisture partitioning for H1, H2, and H5.

6 Conclusions

We study several ways of learning a known moist-physics parameterization within the discretized thermal shallow-water equations (DIMSWE). The study separated three choices: the physical structure imposed on the learned parameterization, the quantity used in the training loss (and hence the necessity of online training), and the extent to which model-generated states are included during training.

A first result is that direct regression of the local moist physics at the state where the parameterization is actually evaluated provides a strong baseline. In M1-Y, the network is trained directly on the pre-moist state $Y^\star = P(X^\star)$. This approach is entirely offline and does not require differentiating through the timestep or recursively evolving the learned model. Nevertheless, the resulting models remain stable over the full DoubleVortex simulation and, particularly for the structured Representations A and B are designed for conservation and reproduce the global evolution accurately.

The one-step H1/M2-Y objective and the recursive H2 and H5 objectives do not provide a uniform improvement over direct regression of M1-Y. For Representations A and B, the state-based models (all except M1-Y) can reduce the later-time error in the phase-change rate A, but this does not imply more accurate recovery of every learned physical process. In Representation B, for example, M1-Y reproduces the analytical rain-production law much more accurately than H1, H2, or H5, even though all four models produce comparatively accurate deployed state trajectories. Thus a learned model can reproduce the resolved state well while representing an individual physical process substantially less faithfully.

The strongest distinction arises from the choice of learned-physics representation. Representation A learns only the phase-change rate and retains both the analytical rain law and the prescribed moist-source structure. Its deployed evolution is robust across all four training objectives. Representation B additionally learns the rain-production rate but retains the known algebraic coupling among the moist tendencies. Its deployed states also remain accurate and total-water conservation is preserved, although the learned rain law is considerably more sensitive to the training objective. Representation C instead learns the four moist-source components independently. The resulting source is not constrained to satisfy the known total-water and thermodynamic identities or the physical two-rate source structure. Although M1-Y produces a comparatively accurate Representation-C model, the H1, H2, and H5 sequence develops progressively larger conservation errors, unphysical cloud–rain partitioning, and negative discrete moisture values.

The recursive calculations provide an additional caution. The H2 and H5 training errors decrease over their respective optimization stages, but for Representation C the error after the full 160-step deployed calculation increases. Improving a two- or five-step training objective therefore does not necessarily improve the much longer deployed trajectory. In the present study, the recursive stages were limited to only 20 optimization iterations and were initialized sequentially from the preceding model, so these results should not be interpreted as a definitive comparison of rollout horizon alone.

All learning strategies also show a substantial difference between errors over the training portion of the DoubleVortex trajectory and the later evaluation portion. The latter is a continuation of the same deterministic flow rather than a new physical regime or initial condition, so the observed degradation already indicates that accurate fitting over a large number of spatial samples does not guarantee temporal generalization as the model state evolves.

Taken together, the results suggest that increasing the complexity of the training objective is not by itself sufficient to improve a learned physical parameterization. For this problem, two simpler considerations are at least as important: training the network on the state at which the parameterization is actually called, and retaining known physical structure in the learned source. Once these are satisfied, direct offline regression can already provide stable and accurate long-time deployment, while one-step and recursive state-based training provide more limited and representation-dependent benefits.

Several extensions remain important. The present calculations use a single DoubleVortex trajectory, a fixed neural-network architecture, and a single initialization. Broader training data, additional flow realizations, larger networks, and longer optimization of the recursive objectives would help separate limitations due to data coverage, model capacity, and optimization. The current networks also do not enforce positivity, and Representations B and C do not impose the analytical sign or threshold structure of the learned rain process. Incorporating such physical constraints, as well as using specific variables such as (h, b, q_v, q_c, B) that more directly expose the analytical moist-physics dependencies, provides a natural direction for future work.

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